

# Feature spreading, redundancy, and blocking behavior

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*Revised July 2026*

## Abstract

We analyze a collection of long-distance assimilation patterns using Boolean Monadic Recursive Schemes (BMRS; Bhaskar et al. 2020, Chandlee & Jardine 2021). BMRS describe input-output maps as logical transductions using conditional IF...THEN...ELSE... syntax. These patterns emerge from varying the presence, type, and position of redundancy conditions within BMRS feature spreading functions. When potential targets fail to receive or propagate harmonic features, redundancy conditions override harmony by blocking feature reception, halting propagation, or preventing both, depending on their type and scope. This offers a unified computational account of variation in feature spreading and blocking without traditional representational assumptions.

## 1 Introduction

Vowel harmony and other ‘long-distance’ assimilation patterns have deservedly enjoyed special attention throughout the history of research in phonological theory. Formal analyses of these types of patterns motivate or otherwise implicate many theoretical assumptions about representation and computation, shaping (and sometimes distorting) our collective understanding of these assumptions as well as of the patterns themselves.

Work in this area has led to the adoption of common terms to refer to key aspects of the analysis of these patterns, terms that often conflate assumptions about representation and computation. Such is the case with the term SPREADING, which we use here to refer to the observation that the output value of a given feature is determined for a subsequence of segments (e.g., all vowels), assimilating with the value specified on a privileged member of that subsequence. ‘Spreading’ implies a particular, autosegmental view of phonological representation and computation (Goldsmith 1976, Clements 1981): a single feature expands its scope, from being associated only with the privileged source to being associated with all members of the subsequence.

Another example is the term BLOCKING, which we use here to refer to the observation that spreading from the privileged source to all members of the subsequence can be interrupted and restarted by some class of segments. ‘Blocking’ also implies the autosegmental view: typically, a member of the interrupting class of segments is specified in such a way that it halts the continued spread of one of the two assimilating feature values.

The autosegmental perspective in turn implies other assumptions about REDUNDANCY, a term we use here to refer to the observation that some feature values of a given segment are predictable based either on other feature values of that segment or on its position relative to other segments. ‘Redundancy’ also comes with representational baggage: predictable feature values are often assumed to be underspecified (whether in the input, in the output, or in both), and to interact in principled ways with spreading and blocking.

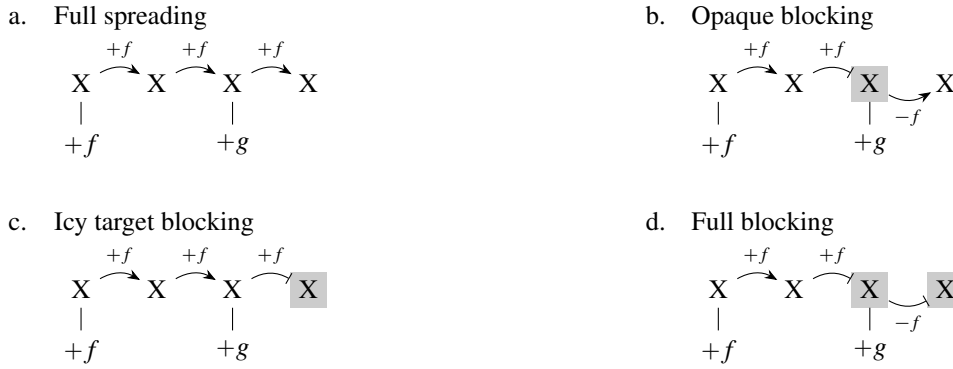
In this article, we show how the formal computational framework of Boolean Monadic Recursive Schemes (BMRS; Bhaskar et al. 2020, Chandlee & Jardine 2021) can be profitably deployed to analyze a significant set of long-distance assimilation pattern types. Even though we continue to use the terms spreading, blocking, and redundancy throughout, their implications about autosegmental representations and underspecification turn out to be entirely dispensable. This result expands upon the *representation as computation* approach to underspecification of Nelson & Baković (2024a,b), and is in the spirit of Baković (2000) and other analyses of long-distance assimilation in Optimality Theory but without the excessive computational expressivity of constraint-based optimization.

BMRS describe phonological input-output maps as logical transductions using conditional IF...THEN...ELSE... syntax, and are limited to subregular expressivity (Bhaskar et al. 2023). As a phonological framework, BMRS have been argued to provide a formalism that can bring together *extensional* computational universals that have been discovered in work rooted in formal language theory (Johnson 1972, Kaplan & Kay 1994, Heinz 2018) and *intensional* substantive universals that have emerged from descriptive and theoretical phonological analysis (e.g. Greenberg 1966,

Hyman 2008). While the BMRS syntax can encode any subregular function, our aim with this work is to identify the computational structures (i.e., extension) and basic predicate types (i.e., intension) that underlie different types of long-distance assimilation patterns.<sup>1</sup> Our proposal thus builds upon recent work that relates different types of phonological generalizations to specific BMRS structures (Chandlee & Jardine 2021, Jardine & Oakden 2025, Nelson & Baković 2024a).

The schematic illustrations in (1) represent the four main types of feature spreading and blocking patterns that we focus on in this article, highlighting the key differences among them. In all four cases, rightward spreading of  $[+f]$  is shown to be initiated by the first element in the string, and the third element is flagged as a potential disruptor of this rightward spread. This is indicated with a  $[+g]$  specification, standing in for whatever property is responsible for this potential disruption. An arc with an arrowhead indicates that the  $[\pm f]$  value succeeds in spreading from one element to another, the failure of spreading where it otherwise might be expected is indicated by a bar at the end of an arc, and the independence of an element from spreading is indicated by shading.

(1) Schematic illustrations of spreading and blocking types



The behavior of the third and fourth elements is what distinguishes each pattern from the others. In (1a), the third element both receives and further propagates the spread of  $[+f]$  despite its in principle incompatible  $[+g]$  specification; for obvious reasons, we refer to this type of pattern as **FULL SPREADING**. In (1b), the third element does not receive the spread of  $[+f]$  and instead initiates its own spread of  $[-f]$ ; this is the familiar type of **OPAQUE BLOCKING** pattern. The third element is shaded in this instance to indicate that its surface value of  $[\pm f]$  is independent of the spreading of  $[+f]$  from its left. In (1c), the third element receives the spread of  $[+f]$  but does not further propagate it. The third element here is dependent on spreading from its left, but the following fourth element is not, and is hence shaded. The third element in such a case has been dubbed an *icy target* (Jurgec 2011), so we refer to this type of pattern as **ICY TARGET BLOCKING**. In (1d), the third element neither receives the spread of  $[+f]$  nor initiates its own spread of  $[-f]$ ; we refer to this less familiar type of pattern as **FULL BLOCKING**. The surface values of both the third and fourth elements in this type of case are independent of the spreading from their left, again indicated with shading.

We derive the four patterns in (1), and variations on them, by altering the presence and scope of **REDUNDANCY CONDITIONS** within a BMRS harmony function. A redundancy condition is an **IF...THEN...** subpart of a larger BMRS function determining the output value of a feature. Some redundancy conditions are **INTERNAL**, preventing association of  $[+f]$  to the segment marked  $[+g]$ , similar to familiar redundancy rules of the form  $[+g] \rightarrow [-f]$ . Others are **EXTERNAL**, preventing association of  $[+f]$  to the segment *neighboring* the one marked  $[+g]$ . The basic idea is that when a potential target does not receive and/or further propagate the spreading feature, it's because there is a redundancy condition that overrides spreading — and it does so in different ways, depending on its nature and scope within the larger harmony function.

The remainder of the article is structured as follows. In §2 we provide an overview of phonological computation with BMRS. We then begin §3 by describing how basic parameters of long-distance assimilation (Archangeli & Pulleyblank 1994) correspond with different parts of BMRS functions, and continue with demonstrations of how each of the four main pattern types in (1) are accounted for by varying the *presence* of internal and external redundancy conditions. In §4 we discuss variants of these patterns that emerge from altering the *scope* of redundancy, and in §5 we broaden our view to include *bidirectional* spreading patterns. In §6 we relate our analyses to broader themes in phonological theories of spreading and blocking, and we conclude in §7.

<sup>1</sup> For slightly different perspectives on the role of computation in describing such patterns, see Aksénova et al. (2024) and Burness et al. (2024).

## 2 Phonological computation as recursive schemes

A BMRS program defines string-to-string mappings via a collection of functions which determine whether a position in an output string carries a given property based on input properties of the string. All of these functions must take as their input a singular domain element (hence *monadic*) and have an output type of `Boolean`  $\in \{\top, \perp\}$ . Functions are built up using `IF...THEN...ELSE...` syntax, defined inductively as follows.

(2) Inductive definition of BMRS functions (Chandlee & Jardine 2021: 492)

- a.  $\top$  (= true) and  $\perp$  (= false) are expressions;
- b. any predicate  $P(t)$  is an expression;
- c. if  $E_1$ ,  $E_2$ , and  $E_3$  are expressions, then so is ‘IF  $E_1$  THEN  $E_2$  ELSE  $E_3$ ’;
- d. nothing else is an expression.

For the purposes of the current work, strings are sequences of phonological segments, delimited on either end by a boundary. The positions that these segments occupy in the string are instances of the variable  $t$  in the predicate  $P(t)$  in (2b). This  $t$  can refer to one of three string positions: the position  $x$  of the domain element under evaluation by the function as a whole, the immediate SUCCESSOR of  $x$ , referred to as  $s(x)$ , or the immediate PREDECESSOR of  $x$ , referred to as  $p(x)$ . The successor and predecessor functions provide a way to refer to the properties of elements in neighboring positions to determine a specific element’s output properties, as is of course common in phonology. Given that each domain element is assumed here to be a segment occupying a unique position in the string, we will henceforth consider these terms to be interchangeable and often collapse their senses, with a bias toward the phonologically familiar term ‘segment’.

Segments are cross-classified by properties, referenced by  $P$  in (2b). Some of these properties are morphological: for example,  $\text{root}(x)$  is  $\top$  (true) if the segment  $x$  is a member of the root, and  $\perp$  (false) otherwise. Other properties are members of the set of phonological features  $\mathcal{F} = \{\text{high}, \text{ATR}, \text{round}, \text{back}, \dots\}$ , interpreted as binary with values ‘+’ and ‘−’ (or, equivalently for our purposes, ‘present’ and ‘absent’). If a given segment  $x$  has a given feature property  $f$  (i.e., if  $f(x) = \top$ ), then it is equivalent to saying that  $x$  is  $[+f]$ , and if it does not have that property (i.e., if  $f(x) = \perp$ ), then it is  $[-f]$ .

Other predicates can be defined based on these basic types. For example, we can define a predicate identifying high unrounded segments as in (3).<sup>2</sup> This function first checks whether a segment  $x$  is  $[+\text{high}]$  ①, and if so, then whether it is also  $[+\text{round}]$  ②. If both ① and ② hold, then  $x$  is high and rounded and so the function evaluates to  $\perp$  ③. If only ① holds, then  $x$  is high and unrounded and the function evaluates to  $\top$  ④. Finally, if ① doesn’t hold, then  $x$  is not high and the function evaluates to  $\perp$  ⑤.

$$(3) \quad \begin{array}{l} [+high] \\ [-round] \end{array}(x) := \text{IF } \overset{\textcircled{1}}{\text{high}}(x) \text{ THEN} \\ \quad \text{IF } \overset{\textcircled{2}}{\text{round}}(x) \text{ THEN } \overset{\textcircled{3}}{\perp} \text{ ELSE } \overset{\textcircled{4}}{\top} \\ \quad \text{ELSE } \overset{\textcircled{5}}{\perp}$$

Predicates may also make reference to domain boundaries, for instance to identify a domain-initial  $x$  via reference to the initial boundary ‘ $\bowtie$ ’ as the predecessor of  $x$  (4a) or a domain-final  $x$  via reference to the final boundary ‘ $\bowtie$ ’ as the successor of  $x$  (4b).

- (4) a.  $\text{initial}(x) := \text{IF } \bowtie(p(x)) \text{ THEN } \top \text{ ELSE } \perp$
- b.  $\text{final}(x) := \text{IF } \bowtie(s(x)) \text{ THEN } \top \text{ ELSE } \perp$

As we’ll see, the ability to define new predicates based on existing ones is a critical feature of the BMRS formalism and its ability to precisely describe phonological patterns.

<sup>2</sup> The conditions and evaluations of this first example function are labeled with circled numbers and cross-referenced in the text to help orient readers who may be new to BMRS. Subsequent functions will be explained in the text but will not be labeled in this way.

## 2.1 Redundancy and faithfulness

The output value of every feature  $[\pm f]$  is determined by a function  $\phi_f(x)$ , defining the conditions under which a segment  $x$  is  $[+f]$  in the output (i.e., where  $\phi_f(x) = \top$ ) and those under which  $x$  is  $[-f]$  in the output (where  $\phi_f(x) = \perp$ ). Consider for example the function  $\phi_{\text{round}}(x)$  in (5), describing a common constraint on segmental inventories: the first line determines that consonants are unrounded, the second line determines that low vowels are unrounded, the third line determines that other (non-low) back vowels are rounded, and the last line determines that other (non-low front) vowels are unrounded.

(5)  $\phi_{\text{round}}(x) :=$  IF  $\text{cons}(x)$  THEN  $\perp$  ELSE  
                   IF  $\text{low}(x)$  THEN  $\perp$  ELSE  
                   IF  $\text{back}(x)$  THEN  $\top$   
                   ELSE  $\perp$

All of the evaluative components of the function in (5) represent (internal) *redundancy conditions*: consonants, low vowels, and front vowels are all redundantly  $[-\text{round}]$ , while non-low back vowels are redundantly  $[\text{round}]$ . The two output values of  $[\pm \text{round}]$  are perfectly predictable from the (input) values of these other features. In terms that may be more familiar to phonologists, a redundancy condition is in relevant respects equivalent to an OT constraint ranking where markedness dominates faithfulness. In the present example, IDENT(round) would be dominated by the following four segmental markedness constraints, with each  $\mathbb{M} \gg \mathbb{F}$  ranking respectively corresponding to each of the lines in (5):  $*[+\text{cons}, +\text{round}]$ ,  $*[+\text{low}, +\text{round}]$ ,  $*[+\text{back}, -\text{round}]$ , and  $*[-\text{back}, +\text{round}]$ .

Now consider three alternative segmental inventories, with the same redundancies as in (5) except that (a) there is a  $[\pm \text{round}]$  contrast among non-low front vowels, (b) there is a  $[\pm \text{round}]$  contrast among non-low back vowels, or (c) there is a  $[\pm \text{round}]$  contrast among all non-low vowels. The  $\phi_{\text{round}}(x)$  function describing each of these inventories would be just like the function in (5), except that the redundancy conditions expressed by the last two lines would be modified in one of the ways shown in (6), respectively.

(6) a. IF  $\text{back}(x)$  THEN  $\top$       b. IF  $\text{back}(x)$  THEN  $\text{round}(x)$       c.  $\text{round}(x)$   
       ELSE  $\text{round}(x)$                       ELSE  $\perp$

The expression in (6a) replaces the final condition in (5), which redundantly specifies non-low front vowels as  $[-\text{round}]$ , with a *faithfulness condition* that instead gives each non-low front vowel the same  $[\pm \text{round}]$  value in the output that it has in the input. The expression in (6b) replaces the penultimate redundancy condition in (5) with a faithfulness condition for non-low back vowels. Finally, the expression in (6c) replaces the entire last two lines of (5) with a blanket faithfulness condition for all non-low vowels. These options are akin to promoting IDENT(round) above one or both of the markedness constraints mentioning  $[\pm \text{back}]$ .

## 2.2 Long-distance effects and recursion

The types of long-distance assimilation patterns that are the focus of this work are described in BMRS with *recursion*: a call from within an output function  $\phi_f(x)$  to itself. For example, the function in (7) progressively and iteratively nasalizes vowels following a nasal consonant.

(7)  $\phi_{\text{nasal}}(x) :=$  IF  $\text{cons}(x)$  THEN  $\text{nasal}(x)$  ELSE  $\phi_{\text{nasal}}(p(x))$

It is worth spelling out in detail how this function determines the output value of  $[\pm \text{nasal}]$  for every segment  $x$  in a string. First, it checks whether  $x$  is a consonant, and if it is, then  $x$  is faithful to its input value of  $[\pm \text{nasal}]$ . This ensures that consonants contrast in terms of  $[\pm \text{nasal}]$ , and do not themselves undergo spreading. If  $x$  is instead a vowel, then it takes on the same value of  $[\pm \text{nasal}]$  as the *output* value of  $x$ 's predecessor. Because this part of the function is recursive, this spreading of  $[\pm \text{nasal}]$  propagates through any uninterrupted sequence of vowels: the first vowel gets its value from the immediately preceding consonant, the next vowel gets its value from the first, and so on.

Note that any domain-initial sequence of vowels — that is, those not ultimately preceded by a consonant — are left undefined by this function. To handle this type of case, we adopt Jardine & Oakden's (2025: 808) "(innocuous) semantic convention", inconsequentially adapted from the original as the *False-at-Edge Convention* (FEC) in (8).

- (8) False-at-Edge Convention (FEC)  
 For any Boolean function  $B$ ,  $B(\{P_s\}(x))$  returns  $\perp$  when it is undefined.

The FEC enables recursive output functions like (7) to terminate when there is no explicit  $\top$  or  $\perp$  element (an output nasal or oral consonant, in this case) to initiate the recursion; the recursive expression effectively ‘spreads’  $\perp$  from the domain edge. We’ll have more to say about the consequences of the FEC in §3.3.

The evaluation table in Figure 1 summarizes how the output value of  $[\pm\text{nasal}]$  is computed by the function in (7). The symbols in the input and output rows are interpreted as follows: ‘N’ is a  $[+\text{nasal}]$  consonant, ‘C’ is a  $[-\text{nasal}]$  consonant, ‘ $\tilde{V}$ ’ is a  $[+\text{nasal}]$  vowel, and ‘V’ is a  $[-\text{nasal}]$  vowel. The ambiguous symbol ‘ $\tilde{V}$ ’ indicates that the  $[\pm\text{nasal}]$  values of vowels in the input is completely irrelevant: the function determines their output values based entirely on spreading from their predecessors.

| Input                    | $\times$ | $\tilde{V}$ | $\tilde{V}$ | N      | $\tilde{V}$ | $\tilde{V}$ | C       | $\tilde{V}$ | $\tilde{V}$ | $\times$ |
|--------------------------|----------|-------------|-------------|--------|-------------|-------------|---------|-------------|-------------|----------|
| $x$                      |          | 1           | 2           | 3      | 4           | 5           | 6       | 7           | 8           |          |
| $\text{cons}(x)$         |          | $\perp$     | $\perp$     | $\top$ | $\perp$     | $\perp$     | $\top$  | $\perp$     | $\perp$     |          |
| $\text{nasal}(x)$        |          | $\times$    | $\times$    | $\top$ | $\times$    | $\times$    | $\perp$ | $\times$    | $\times$    |          |
| $\phi_{\text{nasal}}(x)$ |          | $\perp$     | $\perp$     | $\top$ | $\perp$     | $\perp$     | $\perp$ | $\perp$     | $\perp$     |          |
| Output                   | $\times$ | V           | V           | N      | $\tilde{V}$ | $\tilde{V}$ | C       | V           | V           | $\times$ |

**Figure 1** Evaluation table for iterative progressive nasalization (7). Arrows indicate the flow of computation and source of output values for each segment.

The first block of predicates indicates the input values that are necessary to compute the conditions in the function,  $\text{cons}(x)$  and  $\text{nasal}(x)$ . Segments 3 and 6 are consonants, thus meeting the condition of being  $[+\text{cons}]$  in the input. By the THEN clause of (7), the output  $\phi_{\text{nasal}}(x)$  values of these two consonants are thus dependent on their input  $\text{nasal}(x)$  values, as highlighted by the red arrows in the table. The blue arrows indicate that the output  $\phi_{\text{nasal}}(x)$  value of each of the remaining segments is dependent in part on their satisfaction of the condition of being  $[-\text{cons}]$  in the input. This output value is of course also recursively dependent on the output  $\phi_{\text{nasal}}(x)$  value of the predecessor of each of these segments. Segment 1 lacks a predecessor; the FEC specifies this element as  $[-\text{nasal}]$ , as indicated by the dotted black loop arrow. The output  $[\pm\text{nasal}]$  value of each of the remaining segments (2, 4, 5, 7, and 8) is determined by the output  $[\pm\text{nasal}]$  value of its predecessor, as indicated by the dashed black arrows.

### 2.3 Computing on tiers

Thus far, our discussion of BMRS computation assumes that all functions apply to all segments in a string. For vowel harmony patterns, however, it is standardly assumed that the input-output mapping only pays attention to a subsequence of the string: the (HARMONIC) TIER. Since the representations that BMRS functions compute over are linear structures, some additional work is required to restrict the function to affect only those elements relevant to a given pattern (Nelson & King 2026, Bhaskar et al. 2023). We describe this additional work in the remainder of this subsection.

First, a function partitions the set of segments that are on the tier from those that are not. In (9), the predicates  $E_x$  are stand-ins for expressions that determine either tier membership (9a) or tier non-membership (9b). For example, if  $E_1 = \text{syll}(x)$  then the tier comprises all and only vowels; if  $E_2 = \text{cons}(x)$  then the tier comprises all but consonants.

- (9) a.  $\tau(x) := \text{IF } E_1 \text{ THEN } \top \text{ ELSE } \perp$       b.  $\tau(x) := \text{IF } E_2 \text{ THEN } \perp \text{ ELSE } \top$

Knowing where the tier begins and ends is also important for resolving recursive function calls relativized to the tier. The functions in (10) jointly identify the initial segment of a tier subsequence; identifying the final segment is as simple as replacing ‘ $\times$ ’ with ‘ $\times$ ’ and every instance of  $p(x)$  with  $s(x)$ . The function in (10a) ensures that the property of being string-initial, represented as  $\times(p(x))$  (recall (4a)), spreads covertly from the initial segment up to and including the first tier element, and no further. This information is then referenced by (10b), which evaluates to  $\top$  only for the lone segment  $x$  that both is a member of the tier ( $\tau(x)$ ) and has the (covert) string-initial property  $g(x)$ .

- (10) a.  $g(x) := \text{IF } \times(p(x)) \text{ THEN } \top \text{ ELSE}$   
 $\text{IF } \tau(p(x)) \text{ THEN } \perp \text{ ELSE } g(p(x))$
- b.  $\text{initial}^\tau(x) := \text{IF } \tau(x) \text{ THEN } g(x) \text{ ELSE } \perp$

In (11b–c) we provide an example of a pair of tier-related functions that work together to spread  $[\pm\text{back}]$  from the tier-initial segment to all members of the tier, to be compared with the non-relativized function (11a) by which the string-initial segment directly and iteratively spreads  $[\pm\text{back}]$  rightward through the entire string.

- (11) a.  $\phi_{\text{back}}(x) := \text{IF } \text{initial}(x) \text{ THEN } \text{back}(x) \text{ ELSE } \phi_{\text{back}}(p(x))$
- b.  $\phi_{\text{back}}^*(x) := \text{IF } \tau(x) \text{ THEN}$   
 $\text{IF } \text{initial}^\tau(x) \text{ THEN } \text{back}(x) \text{ ELSE } \phi_{\text{back}}^\tau(p(x))$   
 $\text{ELSE } \perp$
- c.  $\phi_{\text{back}}^\tau(x) := \text{IF } \tau(x) \text{ THEN } \phi_{\text{back}}^*(x) \text{ ELSE } \phi_{\text{back}}^\tau(p(x))$

The function in (11b) relativizes the function in (11a) by wrapping it in a condition that checks whether  $x$  satisfies the property of being on the tier and, if so, allows it to be evaluated by the spreading expression; if it is not on the tier, the function evaluates to  $\perp$  (that is, non-tier elements are redundantly  $[-\text{back}]$ ). We distinguish this relativized version with a star. Note that this embedded spreading expression is identical in structure to (11a), but the first and last predicates are tier-relativized variants.

The first tier-relativized predicate ensures that it is the tier-initial segment that is faithful to its underlying value and that initiates spreading. The last tier-relativized predicate is a call to the function in (11c), which covertly spreads the backness property rightward through the string from those segments on the tier that are independently specified by (11b) — in this case, the faithful tier-initial segment. This covert spreading is then in turn used by (11b) to overtly specify all and only the remaining segments on the tier.

The exact nature of tier projection is currently an open question, interacting directly with the empirical typology of harmony. Many patterns are shaped by certain types of segments behaving *transparently* (i.e., not on the tier) — typically consonants, but often also certain classes of vowels. Space limits our discussion of this issue here, but we hope to explore the effect of transparency on our formal analysis in future work.<sup>3</sup> In what follows we make the simplifying assumption that our functions operate over a projected ‘tier string’ — that is, our functions are defined so that they look like (11a), but we assume that the full implementations of these functions take the form of (11b–c).

## 2.4 Expressivity

Much work over the past decade has explored the expressivity of phonological mappings as they relate to formal function classes (Heinz & Lai 2013, Chandlee 2014, Jardine 2016, Chandlee & Heinz 2018, Chandlee et al. 2018, McCollum et al. 2020, Lambert 2022, Meinhardt et al. 2024, Yolyan 2025). BMRS were originally proposed as a logical characterization of the SUBSEQUENTIAL functions (Bhaskar et al. 2020), which Heinz (2018) proposed as an upper limit for the expressivity of phonological transformations. The subsequential functions can be split into both left and right variants which are related to the direction in which the input string is read into a finite state transducer. When BMRS programs are restricted to only use the successor function ( $s(x)$ ), they are equivalent to the LEFT SUBSEQUENTIAL functions which process strings from left to right. When they are restricted to only use the predecessor function ( $p(x)$ ), they are equivalent to the RIGHT SUBSEQUENTIAL functions which process strings from right to left.

Some of the work cited above has shown that subsequentiality is not sufficiently expressive to capture all known phonological patterns (Heinz & Lai 2013, Jardine 2016, McCollum et al. 2020, Meinhardt et al. 2024), while also being more expressive than necessary to capture most patterns (Chandlee 2014). BMRS is equipped to handle both these types of patterns as well. The class of recursive BMRS programs using both successor and predecessor functions is

<sup>3</sup> In the meantime we point interested readers to Nelson & Baković (2026), who provide a full BMRS analysis of Hungarian backness harmony making crucial use of these types of tier-relativized functions. Furthermore, they show how it is possible to skip both unbounded and a fixed number of segments on the tier.

equivalent to the more expressive class of RATIONAL functions (Bhaskar et al. 2023),<sup>4</sup> while the class of non-recursive BMRS programs is equivalent to the INPUT STRICTLY LOCAL functions.

We limit most of our attention in this article to vowel harmony patterns that can be described with subsequential functions: recursive BMRS programs using only successor or only predecessor. This limitation rules out many conceivable but unattested harmony patterns.<sup>5</sup> For example, so-called ‘majority rule’ patterns, whereby the output value of the harmonic feature depends on which value has greater representation in the input, are known to be generated by OT grammars with common constraint types (Lombardi 1999, Baković 2000), but such patterns fall outside the broader class of rational functions (Eisner 2002, Riggle 2004, Gerdemann & Hulden 2012, Lamont 2021). No BMRS program can compute a majority rule pattern for a simple reason: there is no method available to perform arbitrary counting or numerical comparison.

We also discuss bidirectional spreading patterns in §5, which require greater expressivity than the subsequential functions because they involve recursion using both successor and predecessor functions. This more expressive class of functions includes the debated class of “sour grapes” spreading patterns (Wilson 2003, Jardine 2016, McCollum et al. to appear), but still excludes majority rule patterns.

All of these computational expressivity results are guaranteed by the BMRS formalism. The expressivity of any BMRS program, no matter how many IF...THEN...ELSE... clauses, can only be measured by its recursion using predecessor and/or successor functions. The expressivity of the program as a whole is therefore equivalent to the expressivity of its most expressive function. While description-length characterizations are a common measure of computational complexity (see e.g. Rasin et al. 2021), we eschew them here in favor of expressivity. Description length may be relevant when determining exactly how substantive properties play a role in defining predicates and expressions, but we only indirectly engage with such substantive concerns in this article.

## 2.5 Summary

At its core, BMRS is simply a logical formalism that provides particular computational mechanisms for describing string-to-string functions. This echoes the description of OT as “a formal system in which notions like constraint priority are rigorously defined, but it does not say what the constraints are” (McCarthy 2008: 27). In the same way that OT as a theory of phonology requires a (sub)theory of what is a possible markedness or faithfulness constraint (de Lacy 2011), BMRS also requires a theory of possible predicates and larger expressions separate from the computational system itself in order to be a full working theory of phonology. By focusing on how broad classes of BMRS predicates expressing basic elements of analysis interact to describe attested patterns, we lay the foundation for a more complete theory of spreading and blocking within BMRS.

## 3 Spreading, blocking, and redundancy

We take the basic elements of the analysis of feature spreading and blocking to be as in (12), adapted from Archangeli & Pulleyblank (1994). Given our focus on vowel harmony, we assume throughout that the **domain** or fullest extent of spreading is the phonological word and that the only segments involved in the pattern are vowels (recall §2.3).

(12) Basic elements of spreading and blocking

- a. **Content.** The phonological feature (value) that spreads within the domain.
- b. **Source.** The position in the domain that initiates spreading of the feature.
- c. **Direction.** The orientation of feature spreading, relative to the source.
- d. **Redundancy.** Conditions on the association of the feature with some positions.

Each of these basic elements of spreading and blocking will be encoded in the vowel harmony functions examined in this section, as shown schematically in (13).

<sup>4</sup> This class has somewhat misleadingly been referred to as REGULAR in the literature; the rational functions technically constitute a SUBREGULAR function class. See fn. 1 of both Meinhardt et al. (2024) and Yolyan (2025) for recent corrections of the record on this point.

<sup>5</sup> Given the discussion in §2.3, it is probably more accurate to say that we limit ourselves to the TIER-BASED STRICTLY LOCAL functions (Burness et al. 2021), but this class of functions has yet to be given a BMRS characterization.

(13) Schematic spreading function expressions

- a. **Content determination.**  $\phi_{\text{spf}}(x) := \dots$  : The function determines the output value of the spreading feature  $[\pm\text{spf}]$  for every segment  $x$ .
- b. **Source faithfulness condition.** IF  $\text{src}(x)$  THEN  $\text{spf}(x)$  ELSE  $\dots$  : A segment  $x$  in the source position  $\text{src}$  is faithful to its input  $[\pm\text{spf}]$  value.
- c. **Directional spreading condition.**  $\dots$  ELSE  $\phi_{\text{spf}}(\{p_s\}(x))$  : The output  $[\pm\text{spf}]$  value spreads to a segment  $x$  from its predecessor  $p(x)$  or successor  $s(x)$ .
- d. **Redundancy conditions.**
  - i. **Internal.** IF  $\text{rdn}(x)$  THEN  $\{\top\}$  ELSE  $\dots$  : A segment  $x$  matching a condition  $\text{rdn}$  is redundantly  $[\text{+spf}]$  ( $= \top$ ) or as  $[\text{-spf}]$  ( $= \perp$ ).
  - ii. **External.** IF  $\text{rdn}(\{p_s\}(x))$  THEN  $\{\top\}$  ELSE  $\dots$  : A segment  $x$  whose predecessor  $p(x)$  or successor  $s(x)$  matches a condition  $\text{rdn}$  is redundantly specified as  $[\text{+spf}]$  or  $[\text{-spf}]$ .

All of the spreading pattern functions that we discuss in this article include **content**, **source**, and **direction** expressions (13a–c). Our focus in this section is on the differences among the four types of patterns in (1), which arise from the presence or absence of the different types of expressions that encode **redundancy** (13d).

### 3.1 Full spreading

We begin our illustrations with a vowel harmony pattern with no redundancy conditions, Turkish backness harmony (Clements & Sezer 1982). The feature value assignments that we assume for the vowels of Turkish are given in (14).

(14) Turkish vowels and feature value assignments

|         | [−back]  |          | [+back]  |          |
|---------|----------|----------|----------|----------|
|         | [−round] | [+round] | [−round] | [+round] |
| [+high] | i        | y        | ɯ        | u        |
| [−high] | e        | ø        | ɑ        | o        |

Relevant examples are provided in (15). Roots are indicated with ‘√’ and boundaries between morphemes are indicated with ‘+’. The causative and past tense suffixes both have [+high] vowels and the conditional suffix has a [−high] vowel. The [+high] vowels alternate between [−back] [i] and [+back] [u] and the [−high] vowels alternate between [−back] [e] and [+back] [ɑ], in both cases ultimately depending on the  $[\pm\text{back}]$  value of the preceding root vowel: [−back] in (15a,b) and [+back] in (15c,d).

(15) Turkish backness harmony examples (√root+CAUS+COND+PAST)

- |                                                                                                                 |                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"> <li>a. √gir+dir+sej+di ‘enter’</li> <li>b. √gel+dir+sej+di ‘come’</li> </ol> | <ol style="list-style-type: none"> <li>c. √kur+dur+saj+du ‘break’</li> <li>d. √ağ+dur+saj+du ‘open’</li> </ol> |
|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|

In this case, the **content** (12a) is  $[\pm\text{back}]$ , the **source** (12b) is the (root-)initial vowel, the **direction** (12c) is rightward, and there are no **redundancy** conditions (12d).<sup>6</sup> This FULL SPREADING pattern corresponds to the simplest possible type of BMRS spreading function, which states that the output  $[\pm\text{back}]$  value of a given input vowel  $x$  ( $= \phi_{\text{back}}(x)$ ) is the same as its input value ( $= \text{back}(x)$ ) if  $x$  is initial ( $= \text{initial}(x)$ ); otherwise, it is the same as the output value of  $x$ ’s predecessor ( $= \phi_{\text{back}}(p(x))$ ).

<sup>6</sup> Among [−high, −round] vowels in Turkish, [−back] [e] is redundantly [−low] and [+back] [ɑ] is redundantly [+low], leading these vowels to be harmonically ‘re-paired’ with each other (Baković 2000). We generally set aside these types of redundancies because they are managed by the output functions for the non-spreading feature(s) involved ( $\phi_{\text{low}}(x)$ , in the case of Turkish).



(16) Turkish backness harmony function — full spreading

$$\phi_{\text{back}}(x) := \text{IF initial}(x) \text{ THEN back}(x) \text{ ELSE } \phi_{\text{back}}(p(x))$$

The  $[\pm\text{back}]$  **content** is expressed by the fact that the function determines the output specifications of  $[\pm\text{back}]$  for all vowels. The spreading **source** — that the  $[\pm\text{back}]$  value of all vowels is ultimately determined by (the faithful realization of) the underlying  $[\pm\text{back}]$  value of an initial vowel — is expressed by the IF...THEN... clause of the function: IF the vowel is an initial vowel, THEN its output value of  $[\pm\text{back}]$  is the same as its input value. Finally, the rightward **direction** of spreading is expressed by the ELSE... clause: the output value of any non-initial vowel is the same as the output value of the vowel that precedes it.

### 3.2 Opaque blocking

Adding a redundancy condition at the beginning of the full spreading type of function in (16) results in a function that describes an OPAQUE BLOCKING pattern. We illustrate this with ATR harmony in Standard Yoruba (Awobuluyi 1967). The feature value assignments that we assume for the vowels of Yoruba are given in (17); note the absence of  $[\text{+high}, -\text{ATR}]$  vowels from the inventory, indicated with ungrammaticality asterisks on the relevant symbols.

(17) Standard Yoruba vowels and feature value assignments

|         | [+ATR]  |         | [-ATR]  |         |
|---------|---------|---------|---------|---------|
|         | [+high] | [-high] | [+high] | [-high] |
| [-back] | i       | e       | *ɪ      | ɛ       |
| [+back] | u       | o       | *ʊ      | ɔ       |

(a) --- [+low]  
 (o) --- [-low]

Relevant examples are provided in (18). Examples (18a–f) illustrate full  $[\pm\text{ATR}]$  spreading among  $[-\text{high}]$  vowels. Examples (18g–l) show that  $[\text{+high}]$  vowels do not undergo spreading of  $[-\text{ATR}]$ , and examples (18j–l) show further that  $[\text{+high}]$  vowels initiate leftward spreading of their redundant  $[\text{+ATR}]$  value.

(18) Standard Yoruba ATR harmony examples

|          |                     |           |                |
|----------|---------------------|-----------|----------------|
| a. èrò   | ‘crowd’             | g. ilé    | ‘house’        |
| b. èfó   | ‘vegetable’         | h. ifé    | ‘work’         |
| c. ɛja   | ‘fish’              | i. ilá    | ‘okra’         |
| d. ògèdè | ‘incantation’       | j. òwúrò  | ‘morning’      |
| e. ògèdè | ‘banana/plantain’   | k. òkígbe | ‘magical drug’ |
| f. ɛmèwà | ‘chief’s messenger’ | l. èḡíkà  | ‘shoulder’     |

The **content** in this case is  $[\pm\text{ATR}]$ , the **source** is the final vowel, and the **direction** is leftward. The **redundancy** condition (12d) is that all  $[\text{+high}]$  vowels are  $[\text{+ATR}]$ , so  $[\text{+high}]$  vowels do not undergo  $[-\text{ATR}]$  spreading and block the further propagation of  $[-\text{ATR}]$ . This opaque blocking pattern corresponds to the following BMRS function.<sup>7</sup>

(19) Standard Yoruba ATR harmony function — opaque blocking

$$\phi_{\text{ATR}}(x) := \text{IF high}(x) \text{ THEN } \top \text{ ELSE} \\ \text{IF final}(x) \text{ THEN ATR}(x) \text{ ELSE } \phi_{\text{ATR}}(s(x))$$

The harmony function in (19) states that the output  $[\pm\text{ATR}]$  value of a given input vowel  $x$  ( $= \phi_{\text{ATR}}(x)$ ) is  $[\text{+ATR}]$  ( $= \top$ ) if  $x$  is  $[\text{+high}]$  ( $= \text{high}(x)$ ); that otherwise it is the same as its input value ( $= \text{ATR}(x)$ ) if  $x$  is a final vowel; and that otherwise it is the same as the output value of  $x$ ’s successor ( $= \phi_{\text{ATR}}(s(x))$ ). The redundant  $[\text{+ATR}]$  value of

<sup>7</sup> The  $[\text{+low}, -\text{ATR}]$  vowel is similarly opaque in Standard Yoruba (e.g. [ɛrákpó] ‘type of plant’). This can be analytically accommodated in (19) by adding a redundancy condition with the same top scope as the one for high vowels: IF low( $x$ ) THEN  $\perp$  ELSE ...

both final and non-final [+high] vowels is expressed by the top scope of the internal redundancy condition in the first IF...THEN... clause: IF any vowel is [+high], THEN its output value of [ $\pm$ ATR] is [+ATR].

We have now seen an example of a harmony function without a redundancy condition — full spreading in the form of Turkish backness harmony, (16) — and an example of a harmony function with an internal redundancy condition with top scope — opaque blocking in the form of Standard Yoruba ATR harmony, (19). The two other types of spreading in (1) arise from the addition of an *external* redundancy condition (13d.ii) to each of these types of functions, as we’ll now show.

### 3.3 Icy target blocking

An ICY TARGET BLOCKING pattern involves a set of vowels to which the harmonic feature can spread, but that do not themselves further propagate one of the values of the feature. Jurgec (2011: 91) named these kinds of vowels *icy targets* because they appear to ‘freeze’ spreading. We illustrate icy target blocking with leftward dominant-recessive ATR harmony in Iny (Ribeiro 2002, 2012).<sup>8</sup> The feature values for the (oral, non-low) vowels of Iny are as given in (20).

(20) Iny vowels and feature value assignments

|         | [+ATR]  |         | [−ATR]  |         |          |
|---------|---------|---------|---------|---------|----------|
|         | [+high] | [−high] | [+high] | [−high] |          |
| [−back] | i       | e       | ɪ       | ɛ       | [−round] |
| [+back] | ĩ       | ə       | ĩ       | ə       |          |
|         | u       | o       | ʊ       | ɔ       | [+round] |

The examples in (21) highlight different aspects of the spreading pattern.<sup>9</sup> The examples in (21a–d) show a single instance of leftward spreading of [+ATR] from the final suffix vowel to the root. In each instance, the root contains an icy target — an underlyingly [+high, −ATR] vowel, which undergoes spreading of [+ATR] but freezes its further propagation. Examples (21e–f) both show spreading from a root vowel onto a prefix vowel, and highlight the fact that any [+ATR] vowel initiates strictly leftward spreading, not just a final one. Example (21g) illustrates what happens when there are two input [+ATR] vowels, one final and the other medial: both spread leftward, and moreover, the medial [+high, +ATR] vowel does not behave as an icy target because it is underlyingly [+ATR], as can be appreciated by comparing this example with those in (21b–d).

(21) Iny ATR harmony examples (Ribeiro 2002, 2012)

|    |                   |   |                |                    |
|----|-------------------|---|----------------|--------------------|
| a. | /√krɔbi+dĩ/      | → | [krɔbi'ni]     | ‘a type of monkey’ |
| b. | /√kɔɖu+dĩ/       | → | [kɔɖu'ni]      | ‘a type of turtle’ |
| c. | /√hɛkɔɖi+dĩ/     | → | [hɛkɔɖi'ni]    | ‘oven’             |
| d. | /√kɔlɔkɔ+dĩ/     | → | [kɔlɔko'ni]    | ‘cedar’            |
| e. | /wa+θɛ+√rikɔɛ/    | → | [waθeritɔ'rɛ]  | ‘my sibling’       |
| f. | /r+ɔ+√tʃuhɔ=rɛɾi/ | → | [rotʃu'hɔɾɛɾi] | ‘he is cursing’    |
| g. | /r+ɔ+√tʃuhɔ=r+ɛ/  | → | [rotʃu'hɔɾɛ]   | ‘he cursed’        |

<sup>8</sup> We follow McCollum (2022) and McCollum & Jardine (2022) in using the endonym *Iny* to refer to the language under discussion, as the name previously used in the literature (Karajá) is likely to be a racial epithet. See Ribeiro (2012: 7–8) for relevant discussion. Note that McCollum & Jardine (2022) also pursue a (more comprehensive) BMRS approach to ATR harmony in Iny, arriving at a different analysis than we do here.

<sup>9</sup> The ‘=’ in these examples indicates an enclitic boundary. Note that the [+low] vowel /a/ in example (21e) surfaces faithfully, presumably as [−ATR] and thus disharmonically. This vowel and the [−high] nasal vowels /ã, õ, ã/ are called ‘opaque’ by Ribeiro (2002, 2012), given that they all halt the leftward spread of [+ATR] from a dominant vowel. Only the two [+low] vowels /a, ã/ appear to be opaque in the usual sense, given that they are both (presumably) [−ATR]. The other two [+nasal] vowels /õ, ã/ are both (also presumably) [+ATR], which makes them something more like non-alternating icy targets. We unfortunately lack the space to properly address the issues raised by these facts, and assume that the analysis presented in the text is unaffected by them.

The **content** is  $[\pm\text{ATR}]$ , the **source** is any underlyingly  $[\text{+ATR}]$  vowel, and the **direction** is leftward. Because the harmony pattern is dominant-recessive, any input vowel specified as  $[\text{+ATR}]$  will be faithfully realized as  $[\text{+ATR}]$  in the output with no chance of being overwritten. Icy target behavior is due to an **external redundancy** condition on potential triggers: an input  $[-\text{ATR}]$  vowel is redundantly  $[-\text{ATR}]$  (i.e., spreading is blocked) when it is followed by a  $[\text{+high}, -\text{ATR}]$  vowel. The BMRS function in (22a) fully describes this complex pattern.

(22) Iny ATR harmony function — icy target blocking

- a.  $\phi_{\text{ATR}}(x) := \text{IF } \text{ATR}(x) \text{ THEN } \text{ATR}(x) \text{ ELSE}$   
 $\quad \text{IF } \begin{bmatrix} +\text{high} \\ -\text{ATR} \end{bmatrix}(s(x)) \text{ THEN } \perp \text{ ELSE } \phi_{\text{ATR}}(s(x))$
- b.  $\begin{bmatrix} +\text{high} \\ -\text{ATR} \end{bmatrix}(x) := \text{IF } \text{high}(x) \text{ THEN}$   
 $\quad \text{IF } \text{ATR}(x) \text{ THEN } \perp \text{ ELSE } \top$   
 $\quad \text{ELSE } \perp$

First, the dominant-recessive nature of the harmony is captured by the fact that the predicates for the source and the spreading feature in the first line are identical ( $= \text{ATR}(x)$ ). The second line thus determines the output of underlying  $[-\text{ATR}]$  vowels. Because the external redundancy condition references a  $[\text{+high}, -\text{ATR}]$  vowel, we define a predicate in (22b) to directly reference such vowels in the second line of (22a). If  $x$ 's successor is  $[\text{+high}, -\text{ATR}]$ , then  $x$  redundantly remains  $[-\text{ATR}]$  ( $= \perp$ ). The output  $[\pm\text{ATR}]$  value of all remaining vowels is determined by iterative leftward spreading ( $= \phi_{\text{ATR}}(s(x))$ ).<sup>10</sup>

To underscore our analysis of icy target blocking in Iny, we include a BMRS evaluation table in Figure 2 showing how the output  $[\pm\text{ATR}]$  values of the vowels in  $[\text{h}\epsilon\text{k}\odot\text{d}\tilde{\text{i}}\text{'n}\text{i}]$  ‘oven’ (21c) are computed from the input  $[\pm\text{ATR}]$  values of the vowels in  $/\sqrt{\text{h}\epsilon\text{k}\odot\text{d}\tilde{\text{i}}+\text{d}\tilde{\text{i}}}/$ .

| Input<br>$x$                                                      | $\sqrt{\text{h}}$ | $\epsilon$ | $\text{k}$   | $\odot$ | $\text{d}$ | $\text{i} + \text{d}$ | $\tilde{\text{i}}$ |
|-------------------------------------------------------------------|-------------------|------------|--------------|---------|------------|-----------------------|--------------------|
|                                                                   |                   | 1          |              | 2       |            | 3                     | 4                  |
| $\begin{bmatrix} +\text{high} \\ -\text{ATR} \end{bmatrix}(x)$    |                   | $\perp$    |              | $\perp$ |            | $\top$                | $\perp$            |
| $\text{ATR}(x)$                                                   |                   | $\perp$    |              | $\perp$ |            | $\perp$               | $\top$             |
| $\begin{bmatrix} +\text{high} \\ -\text{ATR} \end{bmatrix}(s(x))$ |                   | $\perp$    |              | $\top$  |            | $\perp$               | $\perp$            |
| $\phi_{\text{ATR}}(x)$                                            |                   | $\perp$    | $\leftarrow$ | $\perp$ |            | $\top$                | $\leftarrow$       |
| <b>Output</b>                                                     | $\text{h}$        | $\epsilon$ | $\text{k}$   | $\odot$ | $\text{d}$ | $\text{i}$            | $\text{'n i}$      |

**Figure 2** Evaluation table for  $/\sqrt{\text{h}\epsilon\text{k}\odot\text{d}\tilde{\text{i}}+\text{d}\tilde{\text{i}}}/ \longrightarrow [\text{h}\epsilon\text{k}\odot\text{d}\tilde{\text{i}}\text{'n}\text{i}]$  ‘oven’. Arrows indicate the flow of computation and source of output values for each segment.

Segment 4 satisfies the source faithfulness condition of being  $[\text{+ATR}]$  in the input. Its output value  $\phi_{\text{ATR}}(x)$  is therefore dependent on its input value  $\text{ATR}(x)$ , as highlighted by the blue arrow. Segment 3 is  $[\text{+high}, -\text{ATR}]$ , meaning that the preceding segment 2 satisfies the external redundancy condition. Its output value is thus by default  $[-\text{ATR}]$ , as highlighted by the red arrows. This is a faithfulness mapping in disguise because only input  $[-\text{ATR}]$  vowels end up being subject to the external redundancy condition, after the source faithfulness condition removes all  $[\text{+ATR}]$  vowels from consideration. The two remaining segments do not satisfy either condition and therefore undergo  $[\pm\text{ATR}]$  spreading from their successors: segment 3's successor is a  $[\text{+ATR}]$  vowel so this value is spread from segment 4; segment 1's successor is a  $[-\text{ATR}]$  vowel, which did not become  $[\text{+ATR}]$  due to the following icy target, so  $[-\text{ATR}]$  spreads from segment 2.

A final comment is in order about the strict unidirectionality of the Iny pattern, which as Ribeiro (2002) notes is typologically unusual especially among dominant-recessive systems. The practical effect of this is intensionally significant. Consider again the examples in (21e–f), which most clearly illustrate strictly leftward spreading: all

<sup>10</sup> Note that ‘THEN  $\text{ATR}(x)$ ’ in the first line and ‘THEN  $\perp$ ’ in the second line of (22a) could be substituted with ‘THEN  $\top$ ’ and ‘THEN  $\text{ATR}(x)$ ’, respectively; the result is an extensionally equivalent function if either or both of these substitutions are made. The formulations of these expressions in (22a) are consistent with parallel ones in previous functions, and with the schematic expressions in (13b).

vowels *after* the dominant [+ATR] vowel surface as [−ATR]. It might appear that this should somehow be attributed to default faithfulness: the only spreading sources are [+ATR] vowels and spreading is strictly leftward, so the only choice these remaining vowels appear to have is to surface with their underlying values intact. But the only default condition in (22a) is the directional spreading condition at the end of the function, ELSE  $\phi_{\text{ATR}}(s(x))$ .

The output [−ATR] values of the final vowels in these examples are thus due to the False-at-Edge Convention (8). The output [−ATR] value of the penultimate vowel in (21e) is then due to spreading from the final vowel, while the values of the penultimate and antepenultimate vowels in (21f) are due to external redundancy and spreading, respectively.

### 3.4 Full blocking

A FULL BLOCKING pattern is one in which a class of vowels neither receives nor propagates the spreading feature. We illustrate this type of pattern with rounding harmony in Halh (Khalkha) Mongolian (Svantesson et al. 2005). The feature values we assume for the short vowels in Halh are given in (23); long vowels are represented as doubled.<sup>11</sup>

(23) Halh Mongolian vowels and feature value assignments

|          | [+ATR]  |         | [−ATR]  |         |
|----------|---------|---------|---------|---------|
|          | [+high] | [−high] | [+high] | [−high] |
| [−round] | i       | e       |         | a       |
| [+round] | u       | o       | ʊ       | ɔ       |

Relevant examples are shown in (24). The examples in (24a–d) illustrate how the [−high] vowel of the past tense suffix alternates to agree in [±round] with the preceding root vowel. The causative suffix in examples (24e–h), on the other hand, is consistently realized with a [+round] vowel, even in examples (24f,h) where it is preceded by a [−round] vowel. The same [−high] vowel of the past tense suffix that was shown to alternate in examples (24a–d) is consistently realized with a [−round] vowel when preceded by the (equally consistently) [+high, +round] vowel of the causative suffix.

(24) Halh Mongolian rounding harmony examples (root+PAST, root+CAUS+PAST)

|    |          |    |             |            |
|----|----------|----|-------------|------------|
| a. | √og+ɣo   | e. | √og+uɣ+ɣe   | ‘give’     |
| b. | √xeeɣ+ɣe | f. | √xeeɣ+uɣ+ɣe | ‘decorate’ |
| c. | √ɔr+ɣo   | g. | √ɔr+uɣ+ɣa   | ‘enter’    |
| d. | √jaw+ɣa  | h. | √jaw+uɣ+ɣa  | ‘go’       |

The **content** is [±round], the **source** is the initial vowel, and the **direction** is rightward. There are two redundancy conditions in this case, one internal and the other external. The **internal redundancy** condition is that [+high] vowels (except for transparent /i/; see fn. 11) are redundantly [−round]. The **external redundancy** condition is that a vowel following a [+high] vowel is redundantly [−round]. The [±round] value of a [+high] vowel is thus neither determined by the preceding vowel nor determinant of following vowels — the hallmark of full blocking. This full blocking pattern can be described with the BMRS function in (25).

<sup>11</sup> Note that rounding and backness are mutually dependent; this is another case of re-pairing (recall fn. 6). The classification of this inventory differs from that in other works, e.g. Goldsmith (1985) and others cited by Svantesson et al. (2005: 222, Appendix E), who note that “[a]ll these assume palatal harmony and are therefore of limited value as analyses of Halh Mongolian.” Svantesson (1985, 2024) and Svantesson et al. (2005) analyze the alternations assumed in these works to be evidence of palatal harmony as *pharyngeal* harmony instead ([±ATR] harmony, in our terms), which we do not address here. Note also that /i/ is transparent to (both [±ATR] and) [±round] harmony in Halh. We assume this to be due to this vowel not being a member of the harmonic tier, as outlined in §2.3.

(25) Halh Mongolian rounding harmony function — full blocking

$$\begin{aligned} \phi_{\text{round}}(x) := & \text{IF high}(x) \text{ THEN } \top \text{ ELSE} \\ & \text{IF initial}(x) \text{ THEN round}(x) \text{ ELSE} \\ & \text{IF high}(p(x)) \text{ THEN } \perp \text{ ELSE } \phi_{\text{round}}(p(x)) \end{aligned}$$

The harmony function in (25) states that the output  $[\pm\text{round}]$  value of a given input vowel  $x$  ( $= \phi_{\text{round}}(x)$ ) is  $[\text{+round}]$  ( $= \top$ ) if  $x$  is a  $[\text{+high}]$  vowel ( $= \text{high}(x)$ ); otherwise, it is the same as its input value ( $= \text{round}(x)$ ) if  $x$  is an initial vowel ( $= \text{initial}(x)$ ); otherwise, it is  $[\text{−round}]$  ( $= \perp$ ) if  $x$ 's predecessor is  $[\text{+high}]$  ( $= \text{high}(p(x))$ ); and otherwise it is the same as the output value of  $x$ 's predecessor ( $= \phi_{\text{round}}(p(x))$ ).

### 3.5 Summary

Each of the four patterns discussed in this section is characterized by whether it includes certain atomic properties: a source faithfulness condition  $S$  which ensures that a vowel  $x$  in the source position is faithful to its input spreading feature value, an internal redundancy condition  $I$  which blocks a vowel  $x$  belonging to a particular class of vowels from receiving (a particular value of) the spreading feature, and an external redundancy condition  $E$  which blocks a neighboring vowel  $\{p\}(x)$  belonging to a particular class of vowels from further propagating the spreading feature to  $x$ . In each case, spreading to  $x$  obtains only when these conditions are *not* met. Assuming that the source faithfulness condition  $S$  is required, there are four possible combinations of the presence/absence of the redundancy conditions  $I$  and  $E$ , each corresponding to one of the four patterns. These combinations and their corresponding patterns are summarized in (26); a graph of the flow of computation for and relationships among the four patterns is given in Figure 3.

(26) Conditions required for the four spreading and blocking patterns

- a. **Full spreading:** only  $S$  is present; spreading to  $x$  obtains only when  $\neg S$  is satisfied (i.e.,  $x$  is not a source).
- b. **Opaque blocking:** both  $S$  and  $I$  are present; spreading to  $x$  obtains only when  $\neg S \wedge \neg I$  is satisfied (i.e.,  $x$  is neither a source nor a blocker).
- c. **Icy target blocking:** both  $S$  and  $E$  are present; spreading to  $x$  obtains only when  $\neg S \wedge \neg E$  is satisfied (i.e.,  $x$  is neither a source nor adjacent to a blocker).
- d. **Full blocking:** all of  $S$ ,  $I$ , and  $E$  are present; spreading to  $x$  obtains only when  $\neg S \wedge \neg I \wedge \neg E$  is satisfied (i.e.,  $x$  is neither a source, a blocker, nor adjacent to a blocker).

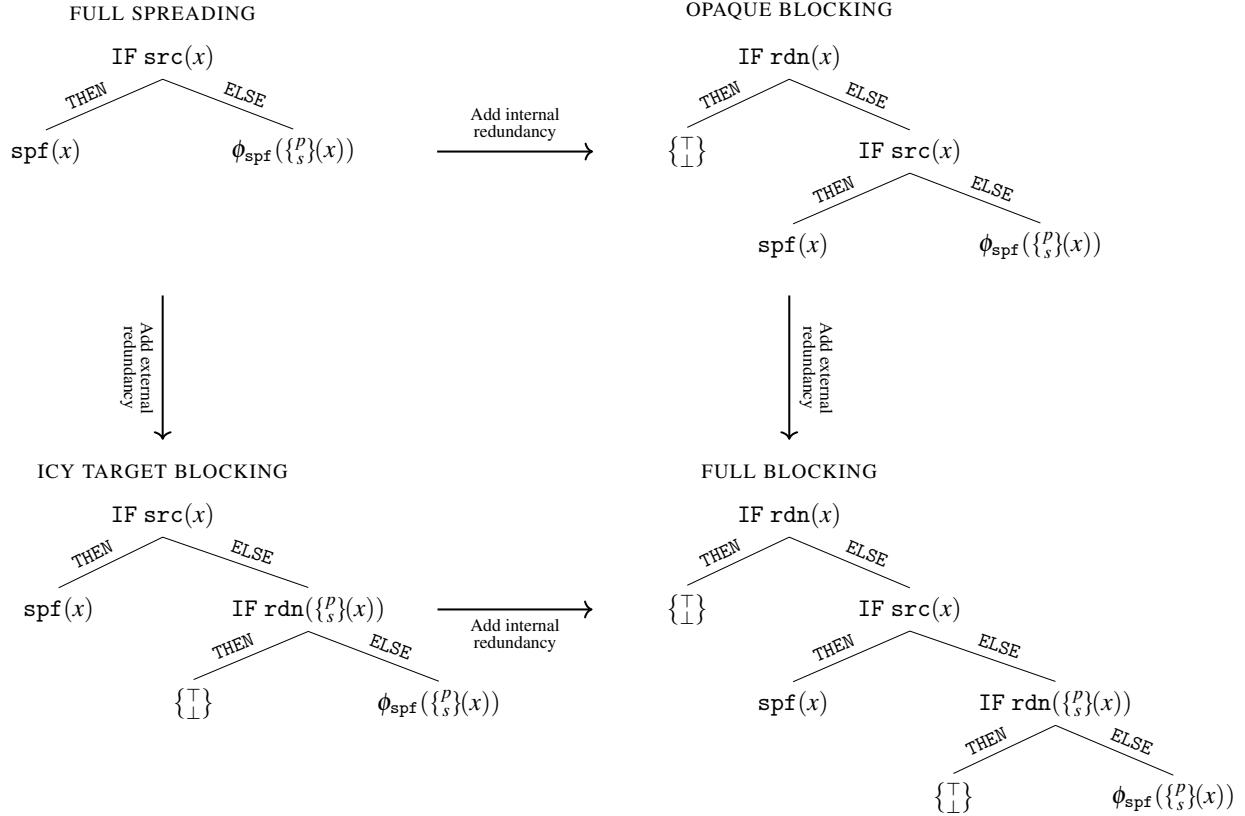
Full spreading is the basic pattern; the only necessary check is whether a vowel is in the source position. Opaque blocking adds a higher-scope internal redundancy condition, because the redundantly specified vowels are unable to undergo spreading but can themselves further propagate the spreading feature. Icy target blocking instead adds a lower-scope external redundancy condition, because this condition still allows a vowel to be the recipient of spreading while disallowing it from further propagating the spreading feature. Finally, full blocking adds both conditions. We have thus shown the relationship among these four types of general spreading and blocking patterns and what appear to be the core elements necessary to compute them.

## 4 Inventory-adjusting patterns

In this section we briefly discuss two additional attested spreading and blocking pattern types that can be described with internal redundancy conditions occupying different positions than those shown in the structures in Figure 3.

### 4.1 Inventory reduction blocking

A different type of opaque blocking is exemplified by Turkish rounding harmony (Clements & Sezer 1982). Relevant examples are provided in (27); (27e–h) are repeated from (15). The  $[\text{+high}]$  vowel of the causative suffix alternates between  $[\text{−round}]$  [i, u] and  $[\text{+round}]$  [y, ʊ], depending on the  $[\pm\text{round}]$  value of the preceding initial vowel:  $[\text{+round}]$  in (27a–d) and  $[\text{−round}]$  in (27e–h). The  $[\text{−high}]$  vowel of the conditional suffix, however, only has the  $[\text{−round}]$  surface reflexes [e, a], blocking the continued spread of  $[\text{+round}]$ . Thus the  $[\text{+high}]$  vowel of the more



**Figure 3** Flows of computation for — and conversion relationships among — full spreading, opaque blocking, icy target blocking, and full blocking patterns.

peripheral past tense suffix still only has the [−round] reflexes [i, u] in (27), in agreement with the redundantly-specified [−round] vowel of the preceding conditional suffix.

(27) Turkish rounding harmony examples (√root+CAUS+COND+PAST)

- |                    |          |                    |         |
|--------------------|----------|--------------------|---------|
| a. √gyl+dyr+sej+di | ‘laugh’  | e. √gir+dir+sej+di | ‘enter’ |
| b. √gør+dyr+sej+di | ‘see’    | f. √gel+dir+sej+di | ‘come’  |
| c. √dur+dur+saj+du | ‘stop’   | g. √kür+dur+saj+du | ‘break’ |
| d. √ol+dur+saj+du  | ‘become’ | h. √atf+dur+saj+du | ‘open’  |

The **content** is [±round], and like backness harmony the **source** is the initial vowel and the **direction** is rightward. There is also an **internal redundancy** condition: [−high] non-initial vowels are redundantly [−round], reducing the set of available vowel types from those available in the source, blocking the potential continued spread of [+round], and initiating [−round] spreading of their own. We refer to this pattern as **INVENTORY REDUCTION BLOCKING** because the inventory of vowels that can surface in the source position is strictly reduced in non-source positions: among [−high] vowels, [+round] [o, ø] are only able to surface in the initial position. Otherwise, the impact of this inventory reduction on spreading is the same as opaque blocking: [+round] spreading is blocked by non-source [−high] vowels because they can only surface as [−round], and subsequent vowels surface as [−round] in agreement with the blocking vowel.

The difference between opaque blocking and inventory reduction blocking lies in the fact that the redundancy condition has top scope in the case of opaque blocking, affecting source and non-source vowels alike, while in the case of inventory reduction blocking the source faithfulness condition has top scope and thus the redundancy condition only affects non-source vowels. Compare the opaque blocking function for Yoruba ATR harmony in (19) with the inventory

reduction blocking function for Turkish rounding harmony in (28a). Because the redundancy condition in the second line of this function requires reference to  $[-\text{high}]$  vowels, we define the predicate in (28b) to directly reference such vowels in the harmony function rather than referencing them more indirectly.

(28) Turkish rounding harmony function — inventory reduction blocking

- a.  $\phi_{\text{round}}(x) := \text{IF } \text{initial}(x) \text{ THEN } \text{round}(x) \text{ ELSE}$   
 $\quad \text{IF } [-\text{high}](x) \text{ THEN } \perp \text{ ELSE } \phi_{\text{round}}(s(x))$
- b.  $[-\text{high}](x) := \text{IF } \text{high}(x) \text{ THEN } \perp \text{ ELSE } \top$

The harmony function in (28a) states that the output  $[\pm\text{round}]$  value of a given input vowel  $x$  ( $= \phi_{\text{round}}(x)$ ) is the same as its input value ( $= \text{round}(x)$ ) if  $x$  is an initial vowel ( $= \text{initial}(x)$ ); otherwise it is  $[-\text{round}]$  ( $= \perp$ ) if  $x$  is  $[-\text{high}]$  ( $= [-\text{high}](x)$ ); and otherwise it is the same as the output value of  $x$ 's predecessor ( $= \phi_{\text{round}}(p(x))$ ). The  $[\pm\text{round}]$  content is expressed by the fact that the function determines the output specifications of  $[\pm\text{round}]$  for all vowels. The initial source of spreading is expressed by the first IF...THEN... clause of the function: IF the vowel is an initial vowel, THEN its output value of  $[\pm\text{round}]$  is the same as its input value. The redundancy of  $[-\text{high}]$  vowels is expressed by the second IF...THEN... clause: IF the vowel is  $[-\text{high}]$ , THEN its output value of  $[\pm\text{round}]$  is  $[-\text{round}]$ . The fact that this redundancy condition only applies to non-initial vowels is expressed by the embedding of this second IF...THEN... clause after the determination of the source, ensuring that every initial vowel is faithful to its input  $[\pm\text{round}]$  value. Finally, the rightward direction of spreading is expressed by the final ELSE... clause: the output value of any non-initial vowel is the same as the output value of the vowel that precedes it.

## 4.2 Inventory expansion spreading

It is also possible for a strictly reduced inventory of vowels to surface in the source position as compared to non-source positions. Because this is effectively the inverse of inventory reduction blocking, we refer to it as INVENTORY EXPANSION SPREADING.<sup>12</sup> Inventory expansion spreading is exemplified by ATR harmony in the Ijeṣa dialect of Yoruba (Bamgboṣe 1967). Relevant examples are provided in (29).

(29) Ijeṣa Yoruba ATR harmony examples

- |                                  |                                    |
|----------------------------------|------------------------------------|
| a. ìgọ 'bottle'                  | d. ılá 'okra'                      |
| b. ulé 'house'                   | e. ọsẹ 'work'                      |
| c. ódiwèrè 'he became a mad man' | f. ọ́dìbàbá 'he became an old man' |

Despite the absence of final  $[\text{+high}, -\text{ATR}]$  vowels, the non-final  $[\text{+high}]$  vowels alternate between  $[\text{+ATR}]$  [i, u] in (29a–c) and  $[-\text{ATR}]$  [ɪ, ʊ] in (29d–f), depending on the  $[\pm\text{ATR}]$  value of the following final vowel. The example in (29f) shows that non-final  $[\text{+high}]$  vowels further propagate the leftward spread of  $[-\text{ATR}]$ .

Like Standard Yoruba, the **content** is  $[\pm\text{ATR}]$ , the **source** is the final vowel, and the **direction** is leftward. The **redundancy** condition is also the same, except that only *final* ( $= \text{source}$ )  $[\text{+high}]$  vowels must be  $[\text{+ATR}]$ . The overall surface vowel inventory is thus the same as the one for Standard Yoruba in (17), except with the ungrammaticality asterisks on the  $[\text{+high}, -\text{ATR}]$  vowels removed. This inventory expansion spreading pattern corresponds to the BMRS function in (30).

(30) Ijeṣa Yoruba ATR harmony function — inventory expansion spreading

- $\phi_{\text{ATR}}(x) := \text{IF } \text{final}(x) \text{ THEN}$   
 $\quad \text{IF } \text{high}(x) \text{ THEN } \top \text{ ELSE } \text{ATR}(x)$   
 $\quad \text{ELSE } \phi_{\text{ATR}}(s(x))$

This harmony function states that the output  $[\pm\text{ATR}]$  value of a given input vowel  $x$  ( $= \phi_{\text{ATR}}(x)$ ) is  $[\text{+ATR}]$  ( $= \top$ ) if  $x$  is both final ( $= \text{final}(x)$ ) and  $[\text{+high}]$  ( $= \text{high}(x)$ ); that it is the same as its input value ( $= \text{ATR}(x)$ ) if  $x$  is

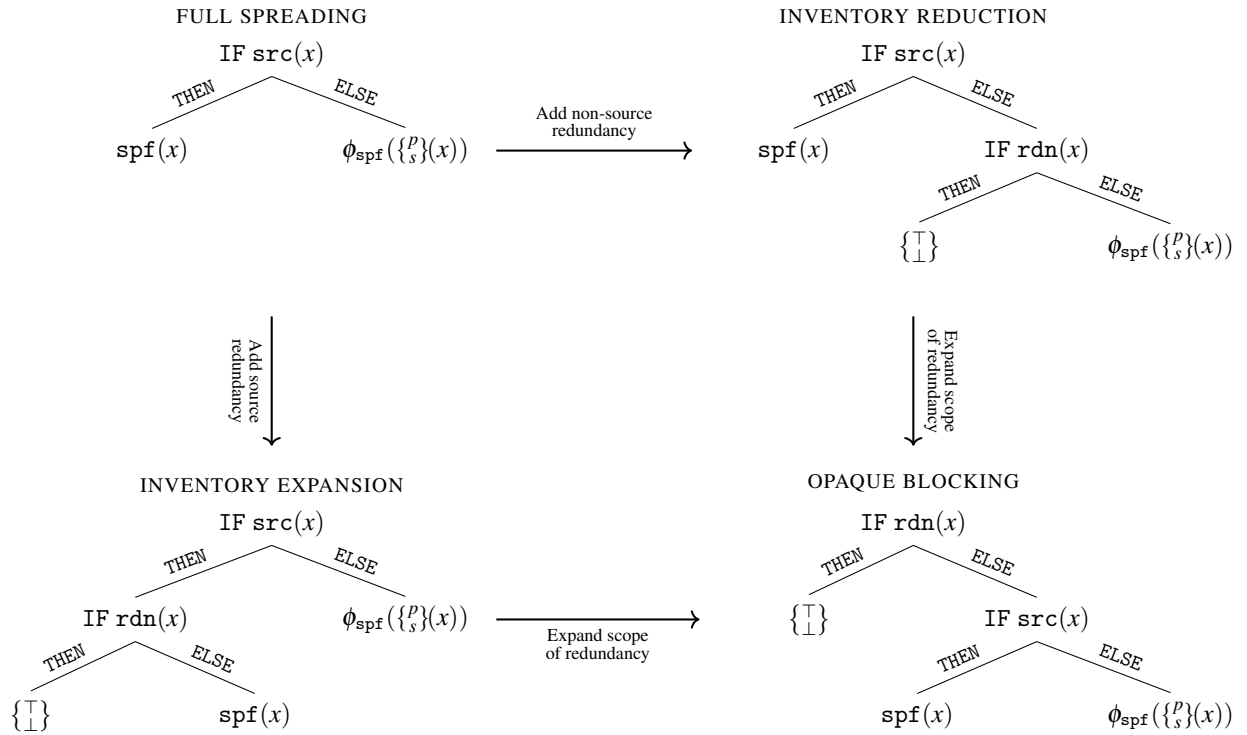
<sup>12</sup> Cases of this type have previously been referred to as *anti-structure preserving* (Baković 2000: 102ff; cf. Kiparsky 1981): a structure preserving process does not derive segment types that are not independently contrastive — or that do not also surface in the conditioning environment of the process, i.e. the source in the case of feature spreading — while an anti-structure preserving process does potentially derive such segments.

otherwise a final vowel; and that otherwise it is the same as the output value of  $x$ 's successor ( $= \phi_{\text{ATR}}(s(x))$ ). The  $[\pm\text{ATR}]$  content of spreading is expressed by the fact that the function determines output specifications of  $[\pm\text{ATR}]$ . The final source of spreading is expressed by the first IF...THEN... clause: IF the vowel is a final vowel, THEN the redundancy and faithfulness conditions apply. The redundancy of final  $[+\text{high}]$  vowels is expressed by the second IF...THEN... clause: IF the final vowel is  $[+\text{high}]$ , THEN its output value is  $[\pm\text{ATR}]$ . The fact that this redundancy condition only applies to final vowels is expressed by the embedding of this second IF...THEN... clause before the otherwise faithful determination of the source. Finally, the leftward direction of spreading is expressed by the ELSE... clause: the output value of any non-final vowel is the same as the output value of the vowel that follows it.

### 4.3 Summary

The differences among opaque blocking, inventory reduction blocking, and inventory expansion spreading lie in the scope of the (internal) redundancy condition. It has top scope in the case of opaque blocking, determining the spreading feature value of redundant source and non-source vowels alike. In the case of inventory reduction blocking it has scope below the faithful determination of source vowels but above the harmonic determination of non-source vowels, thus determining the spreading feature value of redundant non-source vowels alone. In the case of inventory expansion spreading it is embedded within the top THEN... clause, making it only relevant to the source and thus determining the spreading feature value of redundant source vowels alone.

Figure 4 graphs the flow of computation for and conversion relationships among these three spreading patterns and the redundancy-free full spreading pattern. Full spreading can be converted to inventory reduction blocking



**Figure 4** Flow of computation for — and conversion relationships among — full spreading, inventory reduction blocking, inventory expansion spreading, and opaque blocking patterns.

by adding a redundancy condition to the ELSE... clause, affecting only non-source vowels, or it can be converted to inventory expansion spreading by instead adding the redundancy condition to the THEN... clause, affecting only source vowels. Opaque blocking is the result of either adding a top-scope redundancy condition to full spreading or, equivalently, expanding the scope of the redundancy condition. Having a single top-scope redundancy condition is thus extensionally equivalent to having both lower-scope redundancy conditions.

The different types of spreading and blocking that arise from altering the scope of an internal redundancy condition



raises the question whether altering the scope of an *external* redundancy condition has similar effects. In Figure 3 of §3, we gave the external redundancy condition lower scope than the source faithfulness condition in both the icy target blocking and full blocking diagrams, in an effort to make the conversion relationships between connected patterns more visually apparent in the figure. Whether this positioning is necessary for the natural language examples discussed in §3 varies as follows. For Iny, the external redundancy condition must have lower scope because it relies on the higher-scope source faithfulness condition to first remove all of the [+ATR] vowels from consideration. For Halh, the relative scope of the two conditions makes no difference because the external determinant of the redundancy condition is *x*'s predecessor, which is undefined for the initial vowel source and therefore returns  $\perp$  in that case by the FEC (8). We leave it to future research to work out other consequences of altering the scope of redundancy conditions within various types of harmony functions.

## 5 Bidirectional patterns

The recursivity of the spreading condition in a BMRS harmony function allows the computation to be informed by arbitrarily distant information. We have thus far only discussed patterns that require the use of UNIDIRECTIONAL recursive computation: patterns where feature spreading is strictly leftward or strictly rightward. Unidirectional recursion situates the functions describing these patterns within the subsequential class of DETERMINISTIC functions (recall §2.4), for which the output value of the relevant property/feature is determined, for every domain element, by information that is either arbitrarily far to the left or arbitrarily far to the right of the domain element, but not both.<sup>13</sup>

Patterns that require BIDIRECTIONAL recursive computation have received heightened attention in the recent literature. These patterns can be divided into two groups, distinguished by the class of functions required to express them. The more expressive of these two classes is the rational, NON-DETERMINISTIC class. Rational functions are those in which the output of at least some domain elements is determined by information that is arbitrarily far on both sides. Patterns requiring this level of expressivity were dubbed UNBOUNDED CIRCUMAMBIENT by Jardine (2016). Functions in the less expressive class were dubbed WEAKLY DETERMINISTIC by Heinz & Lai (2013).<sup>14</sup> For these functions, the output of some domain elements may be determined by information that is arbitrarily far to the left, and that of other domain elements may be determined by information that is arbitrarily far to the right, but for no domain element can the output be determined by information that is arbitrarily far on both sides. Patterns requiring this level of expressivity were dubbed UNBOUNDED SEMIAMBIENT by Meinhardt et al. (2024).

The distinction between deterministic functions on the one hand, and both weakly deterministic and non-deterministic functions on the other, relates directly to recursive computation within BMRS. If recursion is only used in one direction, then the pattern can be classified as deterministic. If it requires recursion in both directions, then it belongs to one of the more expressive classes. While there are ways to extend the technical BMRS framework to capture bidirectional patterns (Bhaskar et al. 2023, Yolyan 2025), and thus to extend the expressivity of the formalism, we aim to show here that our general characterization of feature spreading and blocking patterns remains intact.

### 5.1 Unbounded semiambience

An example of a bidirectional, full spreading harmony pattern of the unbounded semiambient type is found in the root-controlled ATR harmony pattern of Degema (Keri 1997, Archangeli & Pulleyblank 2007). As the examples in (31) show, both suffix and prefix vowels alternate in  $[\pm\text{ATR}]$  agreement with the root vowel.

(31) Degema ATR harmony examples

- a.  $\bar{u} + \sqrt{f\acute{o}} + \bar{a}$       'the state of being white'
- b.  $\bar{u} + \sqrt{b\acute{o}m} + \bar{a}m$       'beating'
- c.  $u + \sqrt{b\acute{i}} + \bar{o}$       'the state of being black'
- d.  $u + \sqrt{d\acute{e}r} + \bar{o}m$       'cooking'

<sup>13</sup> The determining information may also be local to the domain element, whether in the element itself (by internal redundancy) or in a neighboring element (by external redundancy). This kind of local determination has no impact on the expressivity of a deterministic function.

<sup>14</sup> See also Graf (2016), McCollum et al. (2018, 2020), Hao & Andersson (2019), Lamont et al. (2019), O'Hara & Smith (2019), Smith & O'Hara (2019), Koser & Jardine (2020), Meinhardt et al. (2021, 2024), and Yolyan (2025) for further discussion of this class of functions.

Another example of a bidirectional, full spreading harmony pattern of the unbounded semiambient type is found in the dominant-recessive ATR harmony pattern of Diola Fogy (Sapir 1965, Ringen 1975). As the examples in (32) show, both [+ATR] root vowels (like those in /jitum/, ‘lead away’) and [+ATR] suffix vowels (like that in the ventive suffix /ul/) cause vowels to their left and right to alternate in [±ATR] agreement with them.

(32) Diola Fogy ATR harmony examples

- a. ni+√baj+en+u ‘I caused you to have’
- b. ni+√jitum+en+u ‘I caused you to be led away’
- c. ni+√bəj+ul+u ‘I have for you’
- d. ni+√jitum+ul+u ‘I brought you’

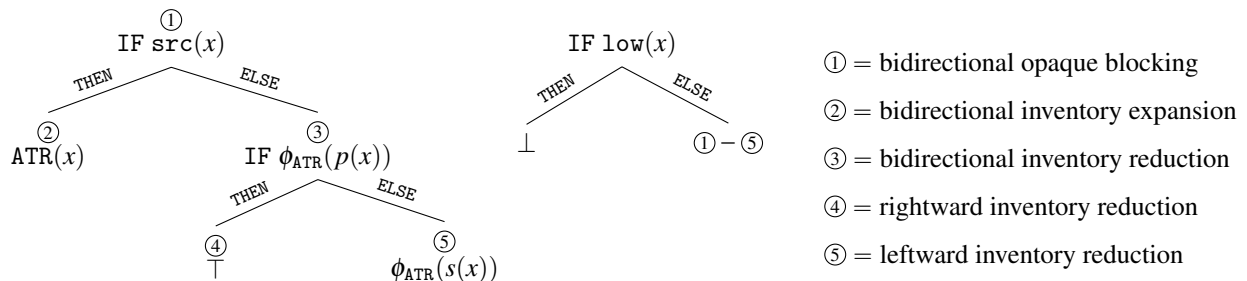
The difference between the root-controlled pattern in Degema and the dominant-recessive pattern in Diola Fogy is the content of the antecedent of the source faithfulness condition,  $\text{src}(x)$ : for Degema  $\text{src} = \text{root}$ , while for Diola Fogy  $\text{src} = \text{ATR}$ .

Note that there is no need to prioritize leftward over rightward spreading or vice-versa, as this choice has no impact on the description of either of these unbounded semiambient patterns (McCollum et al. 2018, Meinhardt et al. 2021, 2024, Yolyan 2025). This is the hallmark of an unbounded semiambient pattern, which can thus be described with a single weakly deterministic function. Two alternative versions for a single BMRS function are provided in (33), following Jardine & Oakden (2025) and Yolyan (2025): one that prioritizes rightward spreading (33a) and another that prioritizes leftward spreading (33b).

(33) Unbounded semiambient [±ATR] harmony functions — full spreading

- a.  $\phi_{\text{ATR}}(x) := \text{IF } \text{src}(x) \text{ THEN } \text{ATR}(x) \text{ ELSE}$   
 $\text{IF } \phi_{\text{ATR}}(p(x)) \text{ THEN } \top \text{ ELSE } \phi_{\text{ATR}}(s(x))$
- b.  $\phi_{\text{ATR}}(x) := \text{IF } \text{src}(x) \text{ THEN } \text{ATR}(x) \text{ ELSE}$   
 $\text{IF } \phi_{\text{ATR}}(s(x)) \text{ THEN } \top \text{ ELSE } \phi_{\text{ATR}}(p(x))$

As before, full spreading can be converted to other pattern types by adding redundancy conditions. Consider for example the results of placing the internal redundancy condition  $\text{IF } \text{low}(x) \text{ THEN } \perp \text{ ELSE } \dots$  in various positions within the function in (33a), as summarized in Figure 5.



**Figure 5** Results (right) of placing an internal redundancy condition (middle) in different positions within an unbounded semiambient harmony function (left). ① all low Vs are [-ATR], ② src low Vs are [-ATR], ③ non-src low Vs are [-ATR], ④ right-of-src low Vs are [-ATR], ⑤ left-of-src low Vs are [-ATR].

Giving the redundancy condition top scope (position ①) describes a pattern where all low vowels are redundantly [-ATR] and thus opaque blockers of [+ATR] spreading in both directions. This matches the description of root-controlled ATR harmony in Akan found in Clements (1981) and Dolphyne (1988), though see now Casali (2012) and Kpogo (2024).

If instead the redundancy condition only has scope over one of the two spreading conditions (positions ④ or ⑤), we describe a pattern with full spreading in one direction and blocking in the other. This matches the description of dominant-recessive ATR harmony in Maasai (Archangeli & Pulleyblank 1994, Baković 2000, Quinn-Wriedt 2013):

low vowels block leftward but not rightward spreading of [+ATR].<sup>15</sup> The three pairs of examples in (34) illustrate both of these aspects of the Maasai pattern.

(34) Maasai ATR harmony examples

- a. ki+√dot+up+ie ‘1PL-pull-MT-APPLIED’
- b. ki+ta+√dot+up+ie ‘1PL-PAST-pull-MT-APPLIED’
- c. i+√duŋ+ifo+re ‘2SG-cut-INTRANS-APPLIED’
- d. i+√as+ifo+re ‘2SG-do-INTRANS-APPLIED’
- e. im+√lipɔŋ+a ‘FEM.PL-female-PROD’
- f. in+√mudoŋ+o ‘FEM.PL-relative-PROD’

In example (34a), both the root and the APPLIED suffix vowels are [+ATR] in the input and there are no low vowels in the form, so all vowels surface as [+ATR]. Adding a low vowel to the left of the leftmost [+ATR] vowel, such as with the PAST prefix in example (34b), blocks leftward spreading. The next pair of examples also illustrates leftward blocking, except that the [+ATR] root in (34c) is replaced with a low vowel root in (34d). The last pair of examples illustrates how low vowels behave differently with respect to rightward spreading: when there are no [+ATR] vowels, as in (34e), all vowels — including the low vowel of the PROD suffix — surface faithfully, but if there are [+ATR] vowels to the left of that low vowel, it undergoes harmony and becomes [+ATR] (34f) — and also undergoes raising and rounding to [o], another instance of re-pairing (recall fn. 6).

The functions responsible for the Maasai pattern are provided in (35). The function in (35a) is responsible for the determination of  $[\pm\text{ATR}]$ . The placement of the internal redundancy condition  $\text{IF low}(x) \text{ THEN } \perp \text{ ELSE } \dots$  coincides with position ⑤ in Figure 5, below rightward spreading but above leftward spreading, thus capturing the directional asymmetry in low vowel behavior. This is thus technically a ‘leftward inventory reduction’ pattern, and were it not for re-pairing, that would actually be the case:  $[\text{low}, +\text{ATR}]$  vowels would surface faithfully in source positions and would be derived in positions to the right of the source, but not in positions to the left of the source. The two functions in (35b,c) disrupt the inventory reduction effect with re-pairing: the redundancy condition  $\text{IF ATR}(x) \text{ THEN } \perp \text{ ELSE } \dots$  affects the determination of both  $[\pm\text{low}]$  and  $[\pm\text{round}]$ .<sup>16</sup>

(35) Maasai ATR harmony and re-pairing functions

- a.  $\phi_{\text{ATR}}(x) := \text{IF ATR}(x) \text{ THEN ATR}(x) \text{ ELSE}$   
 $\text{IF } \phi_{\text{ATR}}(p(x)) \text{ THEN } \top \text{ ELSE}$   
 $\text{IF low}(x) \text{ THEN } \perp \text{ ELSE } \phi_{\text{ATR}}(s(x))$
- b.  $\phi_{\text{low}}(x) := \text{IF ATR}(x) \text{ THEN } \perp \text{ ELSE low}(x)$
- c.  $\phi_{\text{round}}(x) := \text{IF ATR}(x) \text{ THEN } \perp \text{ ELSE}$   
 $\text{IF back}(x) \text{ THEN } \top \text{ ELSE } \perp$

Patterns matching functions with the redundancy condition in positions ② or ③ in Figure 5 are not known to us at this time, nor are bidirectional patterns that can be augmented with external redundancy conditions to achieve icy target blocking or full blocking. Now armed with these analytical possibilities, we look forward to uncovering such patterns.

<sup>15</sup> It also matches the description of emphasis spread in Palestinian Arabic (Davis 1995), where high front segments block rightward but not leftward spread of emphasis/pharyngealization (typically identified as privative [RTR]). See McCarthy (1997) for an OT analysis specifically addressing the ‘process specificity’ of this pattern, and Jardine & Oakden (2025) for a demonstration that a BMRS analysis captures the same process specificity.

<sup>16</sup> An anonymous reviewer asks about “how dependencies are resolved” in a pattern like that in Maasai where “a segment is a target in one direction but not the other”. The input form /a+√duŋ+akm+ie/ ‘3SG-hide-DATIVE-APPLIED’ illustrates: the low vowel of the DATIVE suffix is flanked by [+ATR] vowels, one in the preceding root and the other in the following APPLIED suffix. The function in (35a) unequivocally determines the  $[\pm\text{ATR}]$  value of this low vowel: it will be [+ATR], because the rightward spreading part of the function ( $\text{IF } \phi_{\text{ATR}}(p(x)) \text{ THEN } \top$ ) necessarily takes precedence over the internal redundancy condition part of the function ( $\text{IF low}(x) \text{ THEN } \perp$ ). With re-pairing, this ultimately leads to the right result: [a+√duŋ+okip+ie].

## 5.2 Unbounded circumambience

Tutrugbu has leftward ATR harmony, spreading from the root to prefixes (McCollum & Essegbey 2018, McCollum et al. 2018, 2020). There is also a blocking condition, crucially dependent on information at the other end of the word: a  $[-\text{high}]$  vowel blocks the spread of  $[+\text{ATR}]$  if the initial prefix vowel is  $[+\text{high}]$ . Examples are given in (36).<sup>17</sup>

(36) Tutrugbu ATR harmony examples

- |                        |                                 |
|------------------------|---------------------------------|
| a. bu+tí+√fē           | ‘1P+NEG+√grow’                  |
| b. e+tí+be+√fē         | ‘3S+NEG+FUT+√grow’              |
| c. bũ+tí+√bá           | ‘1P+NEG+√come’                  |
| d. a+tí+ba+√bá         | ‘3S+NEG+FUT+√come’              |
| e. i+dí+√wu            | ‘1S+ITV+√climb’                 |
| f. bũ+ba+di+√wu        | ‘1P+FUT+ITV+√climb’             |
| g. i+tí+ka+á+ba+ba+√wu | ‘1S+FUT+PFV+PRG-VNT+VNT+√climb’ |
| h. e+tí+ke+é+be+be+√wu | ‘3S+NEG+PFV+PRG+VNT+VNT+√climb’ |

This pattern is unbounded circumambient because in order to determine the surface  $[\pm\text{ATR}]$  value of a prefix vowel  $x$  it is necessary to know whether the root vowel, which can be unboundedly far to right of  $x$ , is  $[+\text{ATR}]$ ; whether the initial prefix vowel, which can be unboundedly far to the left of  $x$ , is  $[+\text{high}]$ ; and whether  $x$  itself is a  $[-\text{high}]$  vowel or followed by a  $[-\text{high}]$  vowel, which like the root vowel can be unboundedly far to the right of  $x$ .

Despite requiring information at a potentially unbounded distance in both directions, the Tutrugbu pattern is effectively a special instance of inventory reduction blocking: an internal redundancy condition specifying prefix  $[-\text{high}]$  vowels as  $[-\text{ATR}]$  is embedded below the source faithfulness condition, which allows  $[-\text{high}, +\text{ATR}]$  vowels in the root. What distinguishes this case is the fact that the internal redundancy condition also references additional, non-local information. While we previously invoked external redundancy conditions in situations where the redundancy of an external element blocks spreading, the situation here is crucially different: the internal information (whether the prefix vowel under evaluation is  $[-\text{high}]$ ) is combined with the non-local external information (whether the initial prefix vowel is  $[+\text{high}]$ ) to redundantly specify the  $[-\text{high}]$  prefix vowel under evaluation. The output  $[\pm\text{ATR}]$  values of other prefix vowels are dependent on the output value of their successors, due to the spreading condition.

In order to facilitate the analysis, we define a predicate  $\text{initial/high}(x)$  in (37) which determines whether the initial vowel on a tier string is  $[+\text{high}]$ . This predicate recurses rightward through the input string and ‘spreads’ the information concerning whether the initial vowel of the string is  $[+\text{high}]$  to all vowels to its right.

(37) Tutrugbu rightward-recursive initial  $[+\text{high}]$  vowel predicate

$\text{initial/high}(x) := \text{IF } \text{initial}(x) \text{ THEN } \text{high}(x) \text{ ELSE } \text{initial/high}(p(x))$

The ATR harmony function itself is provided in (38). The first line establishes root vowels as faithful spreading sources. The second line determines whether the domain element  $x$  is  $[-\text{high}]$  — using the same  $[-\text{high}](x)$  predicate as we used for Turkish (28b) — and if so, the third line checks whether the form has an initial  $[+\text{high}]$  vowel, using the rightward-recursive predicate just defined in (37). If this condition is also met, then the  $[-\text{high}]$  vowel is redundantly  $[-\text{ATR}]$  ( $= \perp$ ); otherwise, the  $[-\text{high}]$  vowel agrees with its successor ( $= \phi_{\text{ATR}}(s(x))$ ). Finally, if the antecedent condition isn’t met because  $x$  is  $[+\text{high}]$ , then  $x$  also agrees with its successor. The key to this function describing an unbounded circumambient pattern is that the recursions are CONTRADIRECTIONAL — the input predicate recurses rightward, the spreading function recurses leftward — and that they INTERACT: the leftward-recursive spreading function crucially relies on the information provided by the rightward-recursive input predicate (Heinz & Lai 2013, Jardine 2016, McCollum et al. 2020, to appear, Meinhardt et al. 2021, 2024).

<sup>17</sup> Note that the  $[-\text{high}, +\text{low}, -\text{ATR}]$  vowel  $[a]$  alternates with the  $[-\text{high}, -\text{low}, +\text{ATR}]$  vowel  $[e]$  — yet another instance of ‘re-pairing’.

(38) Tutrugbu ATR harmony function — inventory reduction blocking

$$\begin{aligned} \phi_{\text{ATR}}(x) &:= \text{IF } \text{root}(x) \text{ THEN } \text{ATR}(x) \text{ ELSE} \\ &\quad \text{IF } [-\text{high}](x) \text{ THEN} \\ &\quad \quad \text{IF } \text{initial/high}(x) \text{ THEN } \perp \text{ ELSE } \phi_{\text{ATR}}(s(x)) \\ &\quad \text{ELSE } \phi_{\text{ATR}}(s(x)) \end{aligned}$$

We provide a BMRS evaluation table in Figure 6 to clarify our analysis of Tutrugbu. This table shows how the output  $[\pm\text{ATR}]$  values of the vowels in [bɔbadiwu] are computed from the input  $[\pm\text{ATR}]$  values of the vowels in /bɔ+ba+dɪ+√wu/ (36f).<sup>18</sup> The first block of predicates indicates all of the input values that are necessary to compute the conditions in the actual function, which appear in the next block. Segment 4 satisfies the source faithfulness condition of being a root vowel (and [+ATR], conflated here). Its output value  $\phi_{\text{ATR}}(x)$  is thus dependent on its input value  $\text{ATR}(x)$ , as indicated by the blue arrow. Segment 1 satisfies the condition of being both initial and [+high], so the  $\text{initial/high}(x)$  property is propagated rightward to all elements, as indicated by the red arrows. Segment 2 thus satisfies the internal redundancy condition, meaning that its output  $[\pm\text{ATR}]$  value is by default  $[-\text{ATR}]$ , as indicated by the violet arrows. Finally, the outputs of segments 1 and 3 are dependent on the following vowels' output values for  $[\pm\text{ATR}]$ : segment 3 precedes a [+ATR] vowel, so this value spreads from segment 4; segment 1 precedes a  $[-\text{ATR}]$  vowel, which is redundantly  $[-\text{ATR}]$  due to the internal redundancy condition, so this value spreads from segment 2 to segment 1.

| Input                  | b | ɔ       | + b | a       | + d | ɪ       | + √ w | u       |
|------------------------|---|---------|-----|---------|-----|---------|-------|---------|
| $x$                    | 1 |         | 2   |         | 3   |         |       | 4       |
| root/ATR( $x$ )        |   | $\perp$ |     | $\perp$ |     | $\perp$ |       | $\top$  |
| high( $x$ )            |   | $\top$  |     | $\perp$ |     | $\top$  |       | $\top$  |
| initial( $x$ )         |   | $\top$  |     | $\perp$ |     | $\perp$ |       | $\perp$ |
| $[-\text{high}](x)$    |   | $\perp$ |     | $\top$  |     | $\perp$ |       | $\perp$ |
| initial/high( $x$ )    |   | $\top$  |     | $\top$  |     | $\top$  |       | $\top$  |
| $\phi_{\text{ATR}}(x)$ |   | $\perp$ |     | $\perp$ |     | $\top$  |       | $\top$  |
| Output                 | b | ɔ       | b   | a       | d   | ɪ       | w     | u       |

**Figure 6** Evaluation table for /bɔ+ba+dɪ+√wu/  $\rightarrow$  [bɔbadiwu] (36f). Dashed arrows indicate the flow of computation and source of output values for each segment.

Unbounded circumambient feature spreading patterns have received focused attention in the recent literature. They were effectively claimed to be unattested by Wilson (2003, 2006), but attestations later surfaced: at first with (arguably bounded) metaphony patterns (Walker 2010; cf. Mascaró 2019), then with tonal patterns (Hyman 2011, Jardine 2016), and later with uncontroversially unbounded harmony patterns (like Tutrugbu; McCollum & Essegbey 2018, McCollum et al. 2018, 2020, to appear). What is not in serious dispute is that unbounded circumambient patterns require greater computational expressivity than unbounded semiambient ones (Jardine 2016, Meinhardt et al. 2021, 2024, Yolyan 2025; cf. Lamont et al. 2019, O'Hara & Smith 2019, Smith & O'Hara 2019), and that unbounded circumambient patterns appear to be less commonly attested than unbounded semiambient ones (Jardine 2016, McCollum et al. 2018, 2020, to appear).

We've noted that the Tutrugbu function in (38) is an instance of unbounded circumambient inventory reduction blocking, with the source faithfulness conditions having higher scope than the (recursive) internal redundancy condition. Unbounded circumambient opaque blocking is also predicted to be possible, with the internal redundancy condition having higher scope than the source faithfulness condition, as is inventory expansion spreading, with the internal redundancy condition tucked in the THEN ... clause of the source faithfulness condition. The internal redundancy condition can also be changed to an external one, leading to icy target blocking, or augmented with an external one, resulting in full blocking. The fact that unbounded circumambient patterns are relatively uncommon makes it a particular challenge to identify examples of these predicted patterns, a challenge we hope will be taken up in future research building on this work.

<sup>18</sup> The surface  $[\pm\text{ATR}]$  values of all prefix vowels are predictable, so their input values are arbitrary; we nevertheless specify them underlyingly as  $[-\text{ATR}]$ . McCollum et al. (2020: 222–224) adduce several pieces of evidence for the default value being  $[-\text{ATR}]$  in Tutrugbu.

## 6 Discussion

In this section we situate our approach within broader themes in phonological theory. First, in §6.1, we discuss how our approach offers fundamentally computational solutions to problems that have traditionally been handled representationally. This discussion reveals a natural connection between our approach to feature spreading and blocking to recent BMRS work on underspecification. Then, in §6.2, we discuss the role of substance in our analysis. We argue that substantive properties are ultimately orthogonal to the main points made in this article as they delimit *which types* of segments can act as blockers/undergoers of harmony, while our goal has been to show *how* a segment acts as a blocker/undergoer within a harmony system. Finally, in §6.3, we discuss how our BMRS approach relates to previous analyses. The unifying factor is the existence of a fixed span (or, domain) within which segments all share a feature value in the output.

### 6.1 Representation as computation

The approach to feature spreading and blocking patterns that we have developed in this article re-characterizes what are traditionally thought of as *representational* properties of such patterns as *computational* properties instead. Consider, for example, the traditional representational account of feature spreading and blocking (see e.g. Clements 1981). In this account, what we have here been calling source faithfulness conditions and (internal) redundancy conditions are accounted for with the judicious deployment of feature prespecification and underspecification in the input.

(39) Traditional representational account of source faithfulness and internal redundancy

- a. “To each regular root [i.e., the source position(s)] there corresponds one [spreading] feature, +F or –F. Affixes [i.e., non-source positions] are not specified for those features at any level.”  
(Clements 1981: 57–58)
- b. “OPAQUE segments are just those segments that are already bound to some [value of the spreading feature], before the phonological rules (and the well-formedness condition) apply.”  
(Clements 1981: 61)

The well-formedness condition referenced in (39b), familiar from the autosegmental analysis of tone (Goldsmith 1976), is responsible for associating the spreading feature of the prespecified source and opaque vowels with the underspecified non-source vowels (Clements 1981: 45); opaque vowels block spreading from the source via the ‘association lines do not cross’ clause of this condition. The well-formedness condition is further supplemented with ‘priority clauses’ to prevent opaque vowels from spreading their prespecified feature values to vowels that are otherwise accessible to the source (Clements 1981: 62, 72), and language-particular adjustments must presumably also be made in order to describe the other types of blocking patterns discussed in §§3–4. While the graphical representation of spreading and blocking is striking and compelling, we contend that at least as much if not more of the descriptive and explanatory work is being accomplished by these essentially computational conditions on inputs and association.

Computation is of course ultimately dependent in some way on the representations that it operates over, so we also cannot entirely eschew representational concerns. What makes our approach to feature spreading and blocking computational rather than representational is that the locus of explanation falls squarely within the domain of the computation. Whether the representation consists of *SPE*-like strings of attribute-value feature matrices or of autosegmental/feature-geometric graphs is completely irrelevant to the input-output map computed by our proposed BMRS harmony functions. The only necessary assumption is that there is a ‘skeletal’ string of segments (or root nodes) and means by which to reference both the positions that they occupy and the featural properties that they have (or are associated with).<sup>19</sup> The source faithfulness and redundancy conditions in our BMRS harmony functions handle the behavior of source and opaque vowels without the assumptions in (39) aimed at distinguishing these vowels

<sup>19</sup> As an anonymous reviewer rightly points out, this is not to say that all questions of phonological representation are irrelevant to our concerns. Rather, we shift the locus of explanation to the substantive content and semantic interpretation of feature predicates, as these allow the computation to identify relevant domains and which segments are on a tier. Recent work in computational phonology has been interested in the equivalence of different phonological representation schemes within certain computational expressivity classes (Strother-Garcia 2019, Oakden 2020, Jardine et al. 2021, Nelson 2022, 2024). What has emerged from this work is the view that many types of representation schemes are essentially equivalent and equally capable of distinguishing different elements even with restricted computational power. That being said, Danis (2025) has pointed out that what is missing from this work is a clearer definition of how natural class formation varies across different representational formalisms. Natural class formation is the exact level at which representations matter for our analyses: which types of elements can be distinguished via some type of system for combining predicates into composite classes. This is an important perspective, but we maintain that there is still an infinite number of ways to *represent* any type of natural class system. With that in mind, we reiterate that the results we have shown in this article are agnostic to specific representational choices, which is why we frame our findings purely in terms of computation.

from remaining non-source vowels in the input representation. Moreover, varying the type (internal vs. external) and position/scope of a redundancy condition produces the set of attested spreading and blocking patterns described in §§3–4.

Our approach thus expands upon the one pursued by Nelson & Baković (2024a,b) to what they call UNDER-SPECIFICATION MAPS, for which they use BMRS to obviate the use of representational underspecification. These authors show that underspecification maps share a specific computational structure, making the choice of specification of putatively underspecified segments arbitrary. This arbitrariness implies that the inherently representational device of underspecification is not needed to analyze the relevant patterns, and that it can in fact lead to contradictory representational demands (Nelson & Baković 2024b, building upon observations by McCarthy & Taub 1992).

Nelson & Baković (2024a) discuss Russian voicing as an example case of an underspecification map. The relevant facts are that sonorants are always [+voice], obstruents are [−voice] domain-finally, obstruents contrast in terms of [±voice] when followed by sonorants, and [±voice] spreads regressively in strings of obstruents.<sup>20</sup> A BMRS analysis of this underspecification map is provided in (40). The first line is the (internal) redundancy condition establishing that sonorants (= `sonorant(x)`) are redundantly [+voice] (=  $\top$ ). The remaining lines thus determine the output voicing of obstruents. The second line states that any obstruent followed by a sonorant (= `sonorant(s(x))`) is faithful to its underlying [±voice] value (= `voice(x)`). The last line begins with another redundancy condition establishing that word-final obstruents (= `final(x)`) are [−voice] (=  $\perp$ ); otherwise, [±voice] spreads regressively (=  $\phi_{\text{voice}}(s(x))$ ).

(40) Russian voicing function

$$\begin{aligned} \phi_{\text{voice}}(x) &:= \text{IF } \text{sonorant}(x) \text{ THEN } \top \text{ ELSE} \\ &\quad \text{IF } \text{sonorant}(s(x)) \text{ THEN } \text{voice}(x) \text{ ELSE} \\ &\quad \text{IF } \text{final}(x) \text{ THEN } \perp \text{ ELSE } \phi_{\text{voice}}(s(x)) \end{aligned}$$

Nelson & Baković (2024a: 354) classify this as an underspecification map because it defines the input-output map for the putatively underspecified feature [±voice], it contains a nested conditional BMRS term the upper condition of which (IF `sonorant(x)` THEN  $\top$ ) partitions the set of targets and the lower condition of which (IF `sonorant(s(x))` THEN `voice(x)`) partitions the set of triggers, in both cases removing the class of segments identified by the antecedent of the corresponding redundancy rule in a traditional analysis ([+sonorant]  $\rightarrow$  [+voice]). This analysis thus does not rely on underspecification: sonorants can be freely specified for [±voice] in the input (akin to the *Richness of the Base* hypothesis of OT), and the function in (40) will reliably assign them their predictable output value of [+voice].

This non-reliance on underspecification is significant because, under a traditional representational approach, underspecification maps and opaque blocking patterns employ underspecification in very different ways. This is because the main class of redundantly specified segments in each case exhibit different behaviors: opaque vowels in opaque blocking patterns are triggers but not targets of spreading, while e.g. sonorants in the Russian underspecification map are neither triggers nor targets of anything but the redundancy rule that specifies them as [+voice]. Opaque vowels must thus be underlyingly specified for their redundant spreading feature value<sup>21</sup> while sonorants in Russian must *not* be so specified, lest they trigger spreading of [±voice]. In order for those same sonorants not to be spreading targets, there must be a separate statement that they cannot ever be specified as [−voice] (or that they cannot be specified for *either* value of [±voice] before the redundancy rule that ultimately specifies them as [+voice]).

By contrast, there is no fundamental disparity between opaque vowels and Russian sonorants in our approach: both are redundantly specified via a (top-scope) internal redundancy condition, and hence neither is a spreading target. The fact that opaque vowels but not Russian sonorants are spreading triggers comes down to the content of the (lower-scope) source faithfulness condition: in Russian it is dependent on a neighboring segment of the same class as the internal redundancy condition (i.e., a following sonorant), while in the case of opaque vowels it is not.

To further reinforce the connection between this approach to underspecification maps and our approach to feature spreading and blocking, note that the Russian voicing function in (40) is in many relevant structural respects equivalent to the kind of function that describes a full blocking pattern. This can be appreciated by comparing (40) with the Halh rounding harmony function in (25). In both cases, the first line of the function is an internal redundancy condition: in Russian it redundantly specifies sonorants as [+voice], while in Halh it redundantly specifies high vowels as [+round]. The second line in both cases is a source faithfulness condition: in Russian it ensures that obstruents

<sup>20</sup> We set aside here the predictable voicing of /ts, tʃ, x/ (Halle 1959), the ambiguous sonorant/obstruent behavior of /v/ (Padgett 2002), and the literature on the difference between obstruent and sonorant voicing (Rice 1993).

<sup>21</sup> Or the relevant redundancy rule must apply early, “before the phonological rules (and the well-formedness condition) apply” — recall (39b).

before sonorants are faithful to their underlying  $[\pm\text{voice}]$  value, while in Halh it ensures that initial vowels (i.e., those whose predecessor is a domain-initial boundary; recall (4a)) are faithful to their underlying  $[\pm\text{round}]$  value. The third line in both cases begins with an external redundancy condition: Russian final obstruents (i.e., those whose successor is a domain-final boundary; recall (4b)) are redundantly specified as  $[-\text{voice}]$ , while Halh non-high vowels whose predecessor is a  $[+\text{high}]$  vowel are redundantly specified as  $[-\text{round}]$ . The function ends in both cases with the spreading condition: regressive  $[\pm\text{voice}]$  spreading in Russian, progressive  $[\pm\text{round}]$  spreading in Halh.

Another notable similarity between the Russian and Halh functions is that the relative position of the source faithfulness condition and the external redundancy condition can be reversed in both cases (recall the discussion at the end of §4). At this level of formal analysis, then, the most significant difference between these two functions is the fact that in Russian, internal redundancy and source faithfulness are both conditioned on the same feature ( $[\pm\text{sonorant}]$ ) — this is a key part of what makes this an underspecification map — while in Halh it is internal and external redundancy that are conditioned on the same feature (in that case,  $[\pm\text{high}]$ ). We leave the potential implications of the formal similarities and differences between these two functions to future research.

## 6.2 Form and substance

In §1 we repeated Chandlee & Jardine’s (2021) claim that BMRS provides a way to unite extensional computational universals with intensional substantive universals. That having been said, the approach we have taken in this article makes only indirect reference to substance through the inclusion of only certain types of predicates in harmony functions. As it currently stands, substance plays no direct role in determining the overarching shape of the patterns we discuss. This is ultimately due to the fact that we have focused primarily on how a given segment can act as a source, undergoer, or blocker (i.e., a tier member) within the system rather than on whether specific types of segments can be members of any of these categories. To put it differently, we have assumed that tier membership is given, and worked from there to understand how spreading and blocking emerge from the presence and scope of different types of predicates within BMRS function definitions.

For example, it is commonly observed that segments that block harmony, or those that are transparent to it, are typically not random within a given harmony system. However, we must stress that the properties of the formal system, including representational assumptions, are not the appropriate locus of explanation for this kind of non-randomness. Consider the following example observation, mentioned by an anonymous reviewer: only obstruents have ever been reported to be transparent to nasal harmony. If we take this as a true fact about the world, then it must be *encoded* within the formal phonological system; it cannot be made to *follow* from it. Even if we embraced the types of representational encoding that we dismissed in the previous section, we are still not left with a satisfying explanation as to why certain types of segments are transparent (or opaque, as the case may be) as opposed to others. In other words, if we were to claim that obstruents are transparent to nasal harmony because they have a particular representation (i.e., they are *encoded* differently), then this is simply a restatement of the observed fact.

If one accepts, as we do, that the formal system is not the correct locus of explanation for why some segments but not others are typically transparent or opaque to harmony, then this result is perfectly adequate. BMRS is a sufficiently rich formal system, allowing one to encode any substantive fact that may be necessary. Explanation, then, will have to come from somewhere external to the formal system. This point of view aligns us with at least some of the positions taken within the so-called “substance-free” approach to phonology (Hale & Reiss 2000, Reiss 2017, Blaho 2008). It is not the case that substance plays no role at all in shaping the empirical record; rather, it does so in a way that is separate from being directly placed within the formal system (cf. Blevins 2004).

## 6.3 Relation to OT approaches

In §6.1 we related our analysis to classic representational accounts of autosegmental spreading and underspecification. Here we turn our attention to a general comparison of our analysis with alternative approaches proposed within OT, which does not necessarily impose those same representational assumptions.

The main way in which the BMRS analysis that we have developed throughout this article differs from analyses that have been proposed within the broader OT framework regards the greater expressive power of optimization in OT (recall §2.4). Consider for example an observation made by an anonymous reviewer, that the ranking in (41a) is equivalent to the BMRS function in (41b), because both “can be interpreted as” (41c).<sup>22</sup>

<sup>22</sup> See also, Nelson (2024: 133–138) for a similar example and discussion.



- (41) a. \*NC<sub>o</sub> » \*VOICEDOBSTRUENTCODA » IDENT(voice)  
 b.  $\phi_{\text{voice}}(x) := \text{IF nasal}(p(x)) \text{ THEN } \top \text{ ELSE}$   
      $\text{IF coda}(x) \text{ THEN } \perp$   
      $\text{ELSE voice}(x)$   
 c. “[I]f you’re an obstruent preceded by a nasal, be voiced; otherwise, if you’re a coda, be voiceless; otherwise, be faithful to your underlying voicing.”

This observation depends greatly on constraint type and ranking, however. For example, it does not hold with respect to the ranking in (42), which describes a majority rule assimilation pattern in strings of obstruents (Lombardi 1999, Baković 2000): the obstruents whose value of  $[\pm\text{voice}]$  is least represented in the input are changed in order to ensure agreement; otherwise (i.e., in case of an input ‘tie’), all obstruents are devoiced.

- (42) AGREE(voice) » IDENT(voice) » \*VOICEDOBSTRUENT

Majority rule patterns are beyond the expressive capability of the BMRS formalism (recall §2.4), and therefore no combination of IF...THEN...ELSE... statements is capable of representing this constraint ranking. So, while some constraint rankings can have direct BMRS IF...THEN...ELSE... implementations, not all can.

There are other differences between the two formalisms that we can be given a closer look. Whether expressed with alignment (Kirchner 1993), positional faithfulness (Beckman 1997), licensing (Walker 2005), feature co-occurrence (Baković 2000), or correspondence (Krämer 2003), all OT analyses are ways to determine optimal output forms. While BMRS is also a type of constraint system, it is one based on *satisfaction* as opposed to optimization: when a certain input structure is satisfied, the output will have a specific property. The output-based recursive “spreading” that we use ultimately divides the input strings into various subparts where all the elements within a (tier) subsequence share the same output value because they share a certain set of input properties. This property of our BMRS analysis provides a way to most directly relate to specific OT analyses.

For example, Span Theory (McCarthy 2004) was proposed as a way to approximate the associations borne from autosegmental relations with string representations in OT (cf. Cole & Kisseberth 1994). McCarthy (2004: 3) provides the following definition of a span.

“A featural span is defined as a constituent whose terminal nodes are segments in a contiguous string. There are different spans for each distinctive feature, analogous to the tiers of autosegmental phonology. The partitioning of segments into spans is exhaustive and nonoverlapping: every segment belongs to exactly one span for each distinctive feature [...] Exactly one segment of every span is designated as the span’s head. All of the segments in a span of the feature [F] are pronounced with the head segment’s value for [F] [...] A span can consist of just a single segment, which is necessarily the head, up to all of the segments in a word, of which just one is the head.”

Our analysis, also based on string representations, implements what appear to be exactly featural spans. Source segments mark an initial span, and the heads of other spans are segments that satisfy an internal redundancy condition or whose neighbor satisfies an external redundancy condition. This is essentially the BMRS way of implementing McCarthy’s HEAD(X,[ $\alpha$ F]) constraints, which require any segment with properties X to be the head of a  $[\pm F]$  span. In BMRS, any segment that is neither in the source position nor internally/externally redundant gets its output value for a spreading feature  $[\pm F]$  via the recursive call within the function. This recursive call terminates upon reaching a source or redundant segment (i.e., a *head*). In BMRS, spans are therefore epiphenominal and ultimately emerge from the interaction of source faithfulness, redundancy, and spreading conditions in the function for a harmonic feature.

The other parallel between OT analyses of vowel harmony and our current analysis has to do with the *Richness of the Base* hypothesis (RotB; Prince & Smolensky 1993/2004). We have assumed throughout that BMRS programs are total functions over the set of possible phonological structures defined by a set of primitives (i.e., the set of feature predicates  $\mathcal{F}$ ). This is related (though not identical) to RotB. By assuming a total function, we are committed to inputs being freely specified. Gaps in the surface inventory can thus be explained in one of two ways. First, they could be accidental: a language simply doesn’t have the type of lexical item that would lead to certain surface structures. Second, they could be explained by the grammar, by the structure of the functions within the BMRS program. We take the latter perspective, which aligns more directly with RotB.

In §4.2 we discussed an example of INVENTORY EXPANSION SPREADING from the Ijẹ̀sa dialect of Yoruba. There, it was the case that final  $[+\text{high}]$  vowels must be  $[+\text{ATR}]$ , while non-final  $[+\text{high}]$  vowels alternate in terms of

[ $\pm$ ATR]. Due to the structure of the function in (29), final [+high] vowels can be specified for any value of [ $\pm$ ATR] and still will automatically surface as [+ATR]. The redundancy condition ensuring that these types of segments surface this way makes their underlying values *arbitrary*. This is the BMRS parallel to an OT constraint ranking where \* [+high, -ATR]# is undominated. In both instances, there is no underlying restriction on the [ $\pm$ ATR] specification of final vowels. The surface distribution is completely explained by the shape of the grammar.

## 6.4 Summary

In this section we have followed through on three theoretical consequences of the BMRS approach to feature spreading and blocking proposed in this article. The first consequence concerns the relatively minor role that particularities of representation play in our approach, compared to the relatively more prominent role that representation has played in accounts of feature spreading and blocking since the mid-1970s. We contend that these representational approaches nevertheless require often substantial computational structure both to determine and to interpret those representations, a fact that becomes obvious when the language-particular differences in blocking that form points in the typology we have described are taken into account. This begs the question whether the representational assumptions are independently needed, beyond their graphically satisfying qualities. Leaning into the independently necessary computational tools, as we have here, provides a direct method for deriving the typology.

The second consequence concerns the role of substance in our analysis. We have shown that the role of substance is quite minimal in explaining the macroscopic shape of the patterns we are primarily interested. In this sense, the structures of BMRS functions which lead to the different types of spreading are the *genotype* while specific predicates that can be slotted into different positions within the function are the *phenotypes* which determine the substantive properties involved in specific instantiations of spreading patterns. Both are important for a holistic understanding of the empirical phenomena, but we view our contribution primarily at the more abstract level.

The third consequence has to do with the relationship between our analysis and alternatives in OT. Here, we made three points. First, while some OT constraint rankings can be characterized in terms of BMRS IF...THEN...ELSE... syntax, there are others for which this is not possible. This ultimately stems from the greater expressive power of output-based optimization. Second, we showed how our BMRS analysis of harmony is similar to Span Theory. The primary difference is that ‘spans’ are epiphenomenal in our BMRS analysis, arising from the way that redundancy and faithfulness conditions interact, whereas they are directly optimized for in the Span Theory analysis. Third, we related the assumption of total functions in our analysis to the Richness of the Base hypothesis by highlighting how redundancy conditions can shape surface distributions, allowing us to place no language-particular restrictions on possible input forms.

## 7 Final remarks

In this article we have defined the basic elements of feature spreading and blocking with BMRS — source faithfulness conditions, spreading conditions, and internal and external redundancy conditions — and showed how they can be deployed to generate a structured typology of attested feature spreading and blocking patterns. This demonstration illustrates one way in which a substantive theory of possible structures in BMRS can be developed. First, the relationship between the output property being computed by a harmony function and the redundancy conditions that form a subpart of that function may be restricted by substantive phonetic properties (see e.g. Archangeli & Pulleyblank 1994). Second, we have shown how the presence and scope of redundancy conditions can substantively restrict the way different BMRS structures interact. Both of these play a role in understanding our proposal more broadly.

Existing alternative approaches struggle to describe one or another of the patterns we have covered in this article, consequently failing to characterize them in the structured way that we have here. Representational approaches, which for the past half century have dominated the study of this area of phonology, require a number of additional mechanisms in order to move beyond the classic full spreading / opaque blocking dyad: inventory expansion spreading is anti-structure preserving (Kiparsky 1981); inventory reduction blocking is contrast licensing in privileged (source) positions (Kaplan 2015, Kaplan & Walker 2024); full blocking is due to harmony being limited to a subset of vowels (e.g. to non-high vowels in the case of Halh rounding harmony; Steriade 1979, van der Hulst & Smith 1987, Svantesson et al. 2005, van der Hulst 2018). Icy target blocking has received a number of different analyses involving metrical structure (Jurgec 2011, McCollum 2022), additional local assimilation rules (van der Hulst 2018), and contrast preservation (Ozburn & Leduc 2022).

The computational approach we have taken is largely agnostic to specific representational choices. It is compatible with any theory of representation maintaining a skeletal tier of some kind, thus moving us closer to identifying the components that are required for any system to compute the types of input-output maps that have been discussed throughout this paper. While these components may take on slightly different forms depending on the specific theory they are embedded within, their function at an abstract level remains the same. Our approach provides a vocabulary for explaining *why* further assumptions are needed to describe certain types of patterns within other approaches. For example, the distinction between inventory expansion and reduction is the result of varying the scope of an internal redundancy condition; in order to compute these types of patterns, any theory needs some way to differentiate between how output forms are computed in different positions. There are various ways to accomplish this, at least one of which is needed in any theory if we want to ensure these types of patterns have formal solutions. Our computational characterization therefore has the capacity not only to describe, but also to diagnose at a metatheoretical level.

One issue with this approach to explanation is determining whether the unattestedness of a given predicted pattern is due to formal grammatical factors or some other, external factors (e.g., phonetics, sound change, learnability, etc.). As we have remarked on several times throughout this article, we believe that the interaction between redundancy and faithfulness conditions provides a new roadmap for exploring spreading and blocking patterns in interesting ways. For example, we now have a specific computational property — external redundancy — which seems to be a component of both the typologically rare and unattested patterns. One advantage of novel theoretical approaches is that they are able to shine a light on previously unrecognized properties of linguistic systems. We are optimistic that if the approach we have pursued is embraced, that even more new insights are likely to emerge.

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